

Understanding the Hydrodynamic and Mass Transfer Performance of Newtonian and Non-Newtonian Twin-Liquid Film Flow

Long He, Hanguang Xie, Zhenhua Yao, Ziming Dong, Yuan Zong,* Ling Zhao, and Gance Dai



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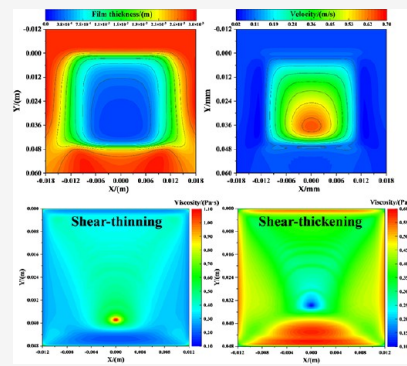
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ABSTRACT: High-viscosity substances and systems are widespread in modern chemical engineering. The twin-liquid film, which combines a wall-bounded film supported by a solid wall and a confined-free film through the opening window, provides an effective way to enhance the interfacial heat and mass transfer of high-viscosity non-Newtonian fluid as well as Newtonian fluid. The average thickness, velocity, and pressure differences of the twin-liquid film are correlated with the effective viscosity (μ_{eff}) and effective Reynolds number (Re_{eff}) unified regardless of fluid rheology. The local differences in the film thickness and velocity between the Newtonian and non-Newtonian twin-liquid films increase with window length, plate thickness, and μ_{eff} . When the diffusion coefficient is inversely proportional to viscosity, the stretching and contraction of the confined-free film with shear-thinning fluid not only boost convective mass transfer but also amplify diffusion, synergistically increasing the mass transfer rate by 5.5 times compared to the solid plate. Twin-liquid films provide a cost-effective approach to enhancing heat and mass transfer efficiency in high-viscosity non-Newtonian systems, such as ionic liquids and nanofluids.



1. INTRODUCTION

High-viscosity substances and systems are prevalent across modern chemical engineering; however, high viscosity is one of the reasons that hinders their applications due to poor fluidity and deteriorated heat and mass transfer performance.

In some inhomogeneous reactors, such as packed columns, falling film tubular heat exchangers, and rotating disc reactors, typical flow patterns involve droplets, rivulets, films, and bubbles. The mass transfer rate $k_L a$ is proportional to fluid viscosity μ with power exponent c , i.e. $k_L a \propto \mu^c$.^{1,2} Deckwer et al.³ measured the value of $k_L a$ in carboxymethyl cellulose (CMC) solutions through oxygen mass transfer experiments in a bubble column, and the power exponent c is -0.85 . This power exponent increases to -0.75 in a sieve column.⁴ This is because the bubbles tend to coalesce more easily with higher viscosity, leading to a decrease in bubble rupture intensity, and subsequently a decrease in mass transfer area and surface renewal frequency.⁵ Comparatively, film flow, which is the common flow pattern in the sieve column and plate type heat exchanger, has the advantages of a high transfer coefficient and large contact area for gas–liquid interfacial heat and mass transfer.^{6,7} The packed column also adopts filmlike flow as the main flow pattern. Song et al.⁸ investigated the effect of liquid viscosity (8 mPa·s $\sim$ 70 mPa·s) on $k_L a$ using 39 types of packings. The results showed that fluid viscosity has a greater impact on the mass transfer coefficient rather than the effective mass transfer area. The power exponent c is -0.4 in the liquid turbulence regime, which is independent of the type of packing.

The rotating disk reactor (RDR) is another film-type reactor that has been widely used in the field of polymer devolatilization. Cheng et al.⁹ measured the value of $k_L a$ by absorbing CO_2 through syrup in an RDR. It was found that as the viscosity of the absorbent increases from 5 Pa·s to 40 Pa·s, the mass transfer coefficient decreased by 70%. Compared to the RDR, the basic flow patterns in a rotating packed bed (RPB) changes from only film flow to the coexistence of liquid film, droplet, and rivulet due to some packings with high specific surface area.¹⁰ The result of the deoxygenation experiment performed by Chen et al.¹¹ in an RPB indicates that $k_L a \propto \mu^{-0.32}$. However, it was found in the RPB that as the viscosity increases from 0.393 to 15.295 Pa·s, the volume fraction of film flow has increased from 30% to 80% while the volume fraction of the liquid droplet and rivulet is less than 20%, which greatly reduces the mass transfer efficiency. Therefore, regardless of the reactor types, a higher viscosity causes trends to form a continuous thick film rather than discrete droplets or bubbles. Compared with the latter, the mass transfer efficiency of the former is lower due to less mass transfer area and suppressed convection. Besides, high viscosity is often associated with non-Newtonian rheological behavior,

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such as in nanofluid,¹² which further pose diverse challenges to heat and mass transfer processes.¹³ Therefore, for highly viscous or non-Newtonian fluids, it is of practical significance to find an efficient way to improve the interfacial heat and mass transfer of liquid films.

Falling film supported by one side wall (wall-bounded film) has been extensively applied in industry, which has inherent advantages such as a large specific surface area, minimal pressure drop, minimal static head, large processing capacity, and low fluid holdup.⁶ Meanwhile, the heat and mass transfer performance of the wall-bounded film with waves has also been investigated. Wasden et al.¹⁴ performed numerical simulations of the mass transfer behavior of this falling film. The results showed that the mass transfer rate of the film flow with large waves is several times higher than that of a waveless film. This is because the generated waves and vortices caused by film instability are efficient mechanisms for generating normal velocity, which is one of the enhancement mechanisms to improve the interfacial mass transfer rate.¹⁵ Levich et al.¹⁶ quantitatively demonstrated that increasing normal velocity by 1% can result in a 15% improvement in the interfacial mass transfer rate. In addition, Alekseenko et al. suggested that film thickness decreases sharply at the trough, thereby reducing the thickness of the local diffusion layer and the diffusion distance. This leads to an improvement in the mass transfer rate.¹⁷ However, most of the previous investigations on the film flow focused on low viscous Newtonian fluids, e.g., water solutions.

As the liquid viscosity increases, film instability is to some extent suppressed, and the film thickness is also found to increase, which all leads to an exponential decrease in the film's mass transfer efficiency. For the non-Newtonian film flow, considering that the stress tensor and strain rate tensor do not follow a proportional relationship, additional nonlinearity is generated into the viscous term. Millet et al.¹⁸ conducted a numerical study based on the Orr-Sommerfeld equation to investigate the flow of shear-thinning fluids down an inclined plane, finding that the shear-thinning effect makes liquid films more prone to instability. Allouche et al.¹⁹ experimentally verified the aforementioned conclusion. Texier et al.²⁰ demonstrated noninertial instability in shear-thickening suspension film flowing down an inclined plane: long surface waves emerge spontaneously at $Re \ll 1$, arising from coupling between free-surface deformation and negatively sloped rheology.²¹ Anderson et al.²² devised a similarity transformation for the film boundary layer problem for non-Newtonian fluids, and an exact solution for accelerating power-law films is thereby presented. It should be noted that theoretical analysis of the film flow with non-Newtonian fluids provides valuable guidance under simple conditions, but it fails for film flows under complex geometric conditions. In recent decades, several CFD methods have been developed to simulate complex film flows. Haeri et al.²³ carried out unsteady 3-D simulations to predict the velocity and thickness of the non-Newtonian falling film flow on inclined plates, and a version of the VOF family of algorithms, CICSAM, is used to track its gas–liquid interface. Sutalo et al.²⁴ measured the film thickness of a power-law fluid and numerically investigated the interface using the CFX-4 code with the VOF method, which considers surface tension by the CSF (Continuum Surface Force) method.

In terms of interfacial mass transfer through films, the study of Newtonian fluids has a history of nearly a hundred years. In 1935, Higbie proposed a solute permeation model based on the mass transfer of smooth laminar film.²⁵ Pigford et al.²⁶ obtained

a solution for the differential equation describing gas absorption into a Newtonian liquid flowing in a falling film with a parabolic velocity profile. However, research about the interfacial mass transfer of the non-Newtonian falling film is relatively rare. Aside from the non-Newtonian rheological behavior, the diffusion coefficient of the gas in a liquid is another important property affecting mass transfer. For some specific situations, the diffusion coefficient can be determined analytically according to the Stokes–Einstein equation. Wasan et al.²⁷ used a short wetted-wall column to study oxygen diffusion coefficients in several aqueous polymer solutions. It is found that in all of the pseudoplastic systems studied, the diffusivity of oxygen is increased with wall shear rate increase and viscosity decrease.

Our previous works have proposed a new type of film flow, the twin-liquid film, which is induced on a plate with opening windows and consists of a wall-bounded film supported by a solid wall and a confined-free film through the opening window. The hydrodynamics of the twin-liquid films have been systematically investigated by experiments and numerical simulations with Newtonian fluid.^{2,28–30} The film thickness, film velocity, surface fluctuation, and internal vortex structure were explored in detail. For low-viscosity Newtonian liquid (0.001–0.01 Pa·s), the simulation results show that the opening window can promote the formation of vortex structures both at the interface and within the film. The wave and vortex of the twin-liquid film disturb and curl the interface concentration boundary layer, which is beneficial for increasing the local mass transfer rate.^{28,30} For medium-viscosity Newtonian liquid (0.04–4 Pa·s), the existence of an opening window increases the velocity of the confined-free film surface velocity to 2.5–16 times and reduces film thickness to 100 μm by adjusting window dimensions. Higher film velocity improves convective mass transfer, and the concentration boundary layer is thinned with the decrease in film thickness, which enhances transverse diffusion. As a result, for medium-viscosity fluid, the maximum local mass transfer rate of the confined-free film can be increased to 2–6.4 times compared with that of the wall-bounded film.²

In various industrial equipment, such as packed columns used in carbon capture technologies and disk reactors for polymer devolatilization processes, studies have demonstrated that opening windows lead to significant enhancements in mass transfer efficiency and flow throughput.^{31–33} However, the underlying enhancement mechanisms of the opening window technique have not yet been fully elucidated. The conceptualization and investigation of twin-liquid films have elucidated the hydrodynamic mechanisms governing the enhancement of fluid flow and mass transfer efficiency in such windowed configurations. Previous investigations on twin-liquid films have been exclusively limited to Newtonian fluids.^{2,34} However, in practical applications, advanced absorbents such as ionic liquids for carbon capture and polymers in devolatilization processes exhibit pronounced non-Newtonian characteristics. Building upon prior research, this study extends the investigation of twin-liquid films from Newtonian to non-Newtonian systems. We systematically elucidate the impacts of non-Newtonian effects (e.g., shear-thickening/thinning) on the hydrodynamic and mass transfer behaviors of twin-liquid films, thereby providing theoretical guidance for the design and optimization of windowed reactors in non-Newtonian fluid processing.

The article is organized as follows: Section 2 details the experimental methodologies for measuring non-Newtonian twin-liquid film thickness. Section 3 describes numerical simulation approaches and validation for non-Newtonian

twin-liquid film systems. Section 4 comprises three subsections: 4.1 delineates fundamental flow patterns of non-Newtonian twin-liquid films; 4.2 analyzes the influences of window geometry and fluid rheological parameters (consistency coefficient and power-law index) on flow field structures; and 4.3 focuses on mass transfer intensification mechanisms. Major conclusions are summarized in the final section.

2. EXPERIMENTAL INVESTIGATIONS

2.1. Materials. The non-Newtonian liquid selected for this study was CMC aqueous solution with mass fractions of 1.38% and 1.91%, respectively. CMC was purchased from Sinopharm Chemical Reagent Co., with a purity greater than 99.5%.

2.2. Physical Properties. The surface tension of the CMC aqueous solution was measured by a BZY-1 automatic surface tension meter (Shanghai Hengping Instrument Factory, China) with a measurement accuracy of 0.0001 N/m. The viscosity of CMC aqueous solutions was measured by a HAAKE MARS60 rotational rheometer (Thermo Fisher Scientific Company, Germany). The shear rate scan ($1\text{--}1000\text{ s}^{-1}$) was conducted using a CP35 cone–plate configuration at $20\text{ }^{\circ}\text{C}$. The test processes were carried out under a dry nitrogen atmosphere to prevent the CMC aqueous solution from absorbing surrounding water during the tests. The measurement was repeated three times, and the average value was taken as the final result. The rheological relationships, i.e., viscosity μ versus shear rate γ , for the CMC aqueous solution agree well with the power-law model, as shown in Figure 1 and Table 1:

$$\mu = K\gamma^{n-1} \quad (1)$$

where K is the consistency coefficient, and n is the power-law index.

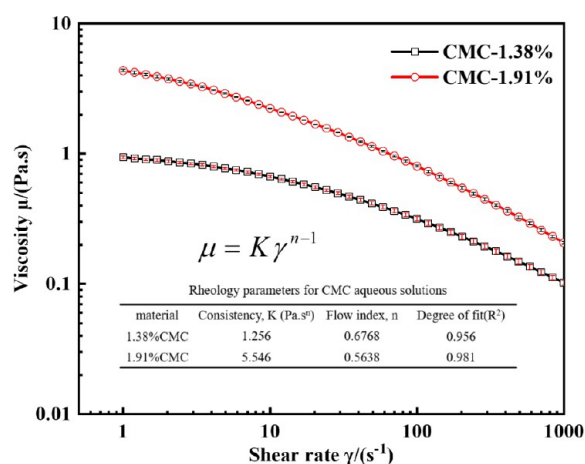


Figure 1. Shear viscosity of CMC aqueous solution with different shear rate at $20\text{ }^{\circ}\text{C}$.

Table 1. Rheology Parameters for CMC Aqueous Solutions
($\mu = K\gamma^{n-1}$)

Material	Consistency, K (Pa.s) ⁿ	Flow index, n	Degree of fit (R^2)
1.38%CMC	1.256	0.6768	0.956
1.91%CMC	5.546	0.5638	0.981

2.3. Film Flow Experiments. The experimental setup is shown in Figure 2. The experiment was carried out at room temperature ($20\text{ }^{\circ}\text{C}$) and atmospheric pressure. A 600×600

mm aluminum plate with a thickness of 1 mm was used as the test plate. The window opening was positioned 300 mm below the liquid inlet. The window dimensions (24 mm in length and 48 mm in width) were fabricated via laser cutting to ensure precision and minimize burrs. The test plate was vertically fixed to an external frame. CMC solutions with mass fractions of 1.38% and 1.91% were employed as the test fluids. A gear pump delivered the liquid to a buffer tank, after which the liquid was released through a flow guide plate into two overflow channels of identical height and introduced to the top of the test plate. The volumetric flow rates on both sides of the plate were regulated by two independent valves and monitored in real time by using gear flowmeters to ensure equal flow rates on both sides. The liquid exiting the bottom of the plate was collected in a trough and returned to the storage tank. Before each experiment, the plate surface was cleaned with a concentrated NaOH solution for prewetting treatment. By measuring the liquid film thickness at different positions of the vertical plate, we ensured the liquid film was uniform on the front and back of the plate and in the transverse direction.

A spectral confocal displacement sensor (CHROCODILE S, Precitec Optronic GmbH, Germany) with an accuracy of $1\text{ }\mu\text{m}$ and a frequency of 1000 Hz was used to measure film thickness in the present work. The sensor enabled continuous monitoring of thickness variations over extended periods. The optical sensor is a device that establishes the corresponding relationship between distance and wavelength through the principle of optical dispersion and uses a spectrometer to decode spectral information to obtain position information. To ensure the light source was focused on the film, the optical probe was fixed on a mobile platform. The mobile platform was fixed on a guide rail that could move along the X direction (incident direction) and Y direction (flowing direction). The movement accuracy of the guide rail was 0.05 mm. The liquid film thickness δ was measured every 1 mm along $X = 0$ and $Y = 0.024\text{ m}$. Before data acquisition, continuous online monitoring of film thickness variations over time was conducted until the flow stabilized. Subsequently, measurements were taken at each monitoring point at a frequency of 1000 Hz for 15 s, resulting in 15,000 data points collected per point. The average film thickness $\bar{\delta}$ at this point is defined as

$$\bar{\delta} = \frac{\sum_{i=1}^N \delta_i}{N} \quad (2)$$

The standard deviation β is defined as

$$\beta = \sqrt{\frac{\sum_{i=1}^N (\delta_i - \bar{\delta})^2}{N - 1}} \quad (3)$$

For the falling laminar film of power-law fluid along the vertical wall, neglecting any apparent slip at the wall, the flow process can be described by the following equations:

$$\rho g - \frac{d\tau}{dz} = 0 \quad (4)$$

where τ is shear stress (Pa). According to the boundary condition, at the gas–liquid interface $z = \delta$ and $\tau = 0$:

$$\tau = \rho g z \quad (5)$$

The power-law constitutive equation for a non-Newtonian fluid is

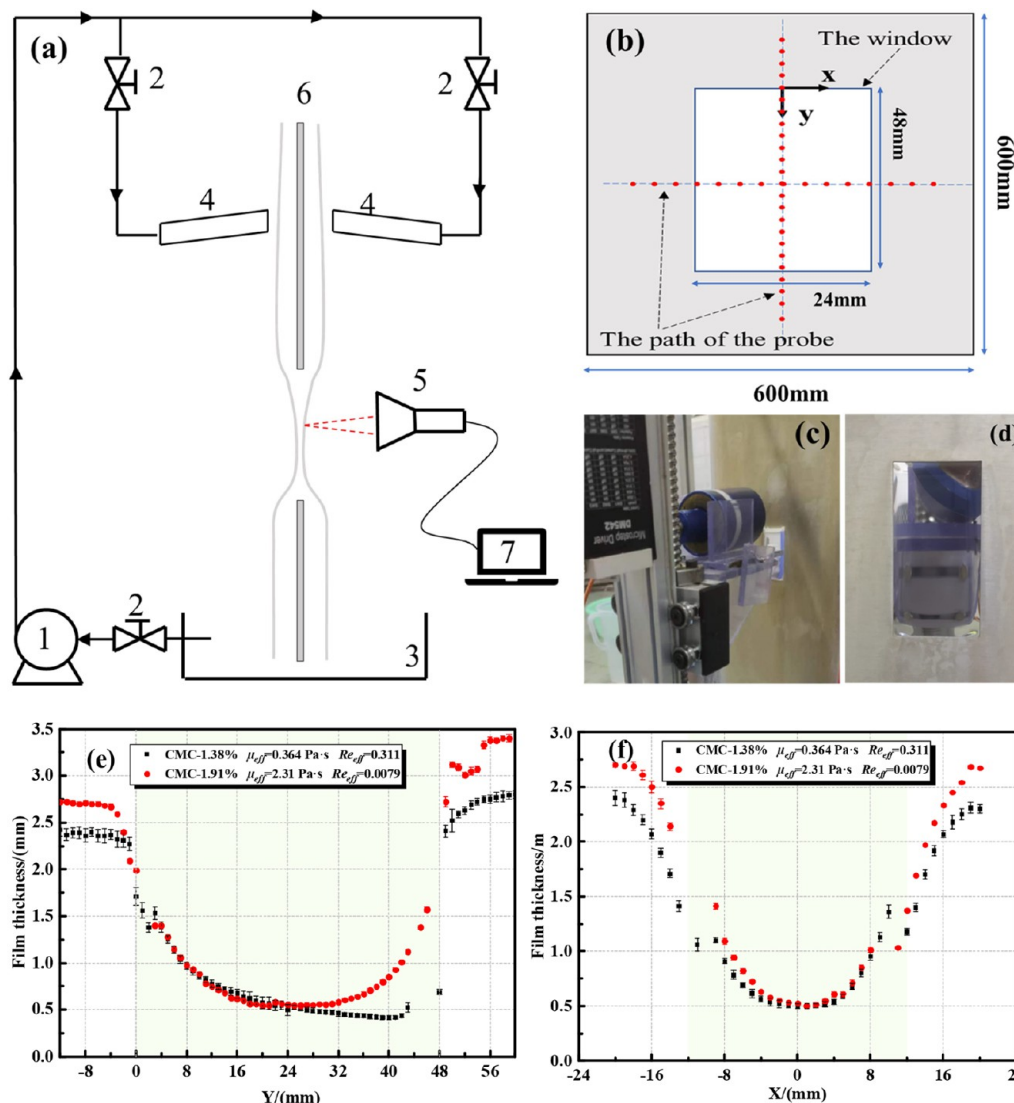


Figure 2. (a) Experimental setup: 1- gear pump; 2- ball valve; 3- Liquid reservoir; 4- liquid distributor; 5- spectral confocal displacement sensor; 6- test aluminum plate; and 7- computer; (b) sketch map for the path of the probe; (c) guide rail; (d) view of the twin-liquid film around the window; (e) film thickness distribution along Y direction at X = 0; (f) film thickness distribution along X direction at Y = 0.024 m. The shaded part in figure (e) and (f) represents the locations of window and same with the following figures.

$$\tau = -K \left| \frac{dU}{dz} \right|^{n-1} \frac{dU}{dz} \quad (6)$$

Combining this with the boundary conditions of $z = 0$ and $U = 0$, the velocity distribution in the liquid film can be obtained:

$$U = \left(\frac{\rho g \delta}{K} \right)^{1/n} \frac{n\delta}{n+1} \left[1 - \left(\frac{\delta - z}{\delta} \right)^{n+1/n} \right] \quad (7)$$

Thus, the liquid film thickness δ is

$$\delta = \left(\frac{K}{\rho g} \right)^{1/2n+1} \left(\frac{2n+1}{n} Q \right)^{n/2n+1} \quad (8)$$

where Q is the volume flow rate per unit plate width (m^2/s).

The average velocity is given by

$$\bar{U} = \left(\frac{1+n}{1+2n} \right) \left(\frac{\rho g \delta}{K} \right)^{1/n} \frac{n\delta}{n+1} \quad (9)$$

Thus, τ can be expressed as

$$\tau = \rho g \delta = K \left[\frac{(1+2n)U}{n\delta} \right]^n \quad (10)$$

In fluid mechanics, the combination of a dynamic pressure head ($1/2\rho U^2$) and a friction resistance coefficient λ is typically used to quantify mechanical energy losses caused by viscous resistance. For liquid film flow, the viscous resistance primarily originates from the wall shear stress τ . By following the computational relationship between the friction resistance coefficient and the Fanning friction factor in laminar flow within circular pipes, the friction resistance coefficient λ for liquid film flow can be derived:

$$\lambda = \frac{8\tau}{\rho U^2} = \frac{24}{Re_{\text{eff}}} \quad (11)$$

where Re_{eff} is the effective Reynolds number, which is expressed as

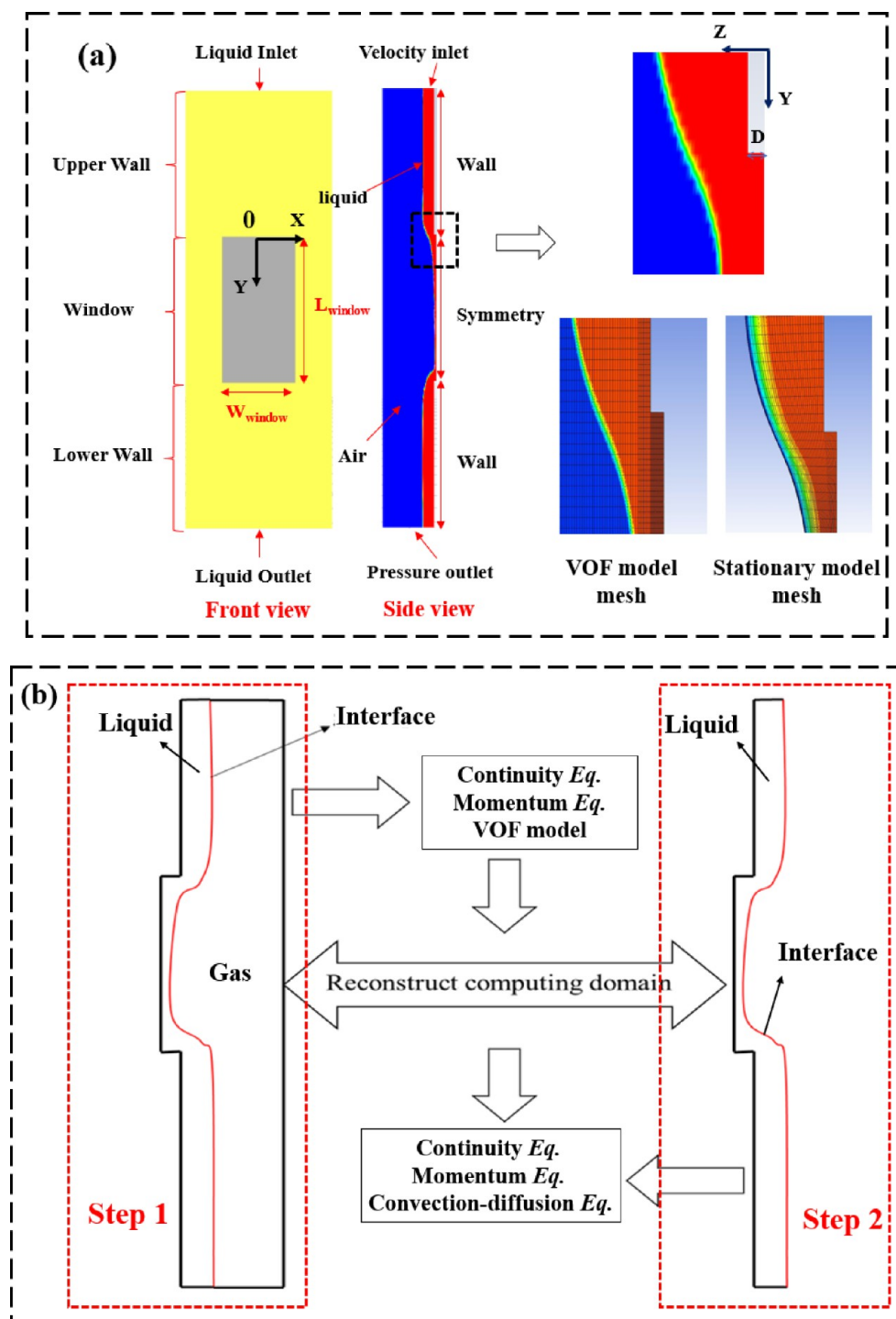


Figure 3. (a) The computational domain and boundary conditions for numerical simulation; and (b) solution strategy and steps for mass transfer simulation.

$$Re_{\text{eff}} = \frac{3\rho\delta^n}{KU^{n-2}\left(\frac{1+2n}{n}\right)^n} \quad (12)$$

When $n = 1$, $Re_{\text{eff}} = \frac{\delta U \rho}{\mu}$, which is consistent with a Newtonian fluid. According to $Re_{\text{eff}} = \frac{\delta U \rho}{\mu_{\text{eff}}}$, the effective viscosity for the power-law non-Newtonian fluid is defined as follows, which considers the viscous force:

$$\mu_{\text{eff}} = \frac{KU^{n-1}\left(\frac{1+2n}{n}\right)^n}{3\delta^{n-1}} \quad (13)$$

From eq 13, it can be observed that the effective viscosity includes the influence of operating conditions (U , δ) and non-Newtonian properties (K , n) on flow resistance.

2.4. Experimental Results and Discussion. For the CMC aqueous solution used in the experiment, the effective viscosity (μ_{eff}) was about 0.364 Pa·s (1.38 wt %) and 2.31 Pa·s (1.91 wt %), and the generalized Reynolds number (Re_{eff}) was 0.311

(1.38 wt %) and 0.0079 (1.91 wt %). Hence, the film flow with these CMC aqueous solutions was in a laminar state; therefore, the film was flat and had no obvious waves, as shown in Figure 2d. The measurement results of film thickness for the plate with a window of size 24 mm × 48 mm are shown in Figure 2e,f. The difference between the measured value and the theoretical value from eqs 7 and 8 of the liquid film thickness on the upper wall ($Y < 0$) was less than 3.5%. Near the windowsill (i.e., $-3 \text{ mm} < Y < 3 \text{ mm}$), several data points were missing. This was mainly attributed to the liquid ridges generated by surface tension and the dramatic change in the film thickness, which caused the laser probe to go out of focus. Flowing into the window region, the film thickness decreased from 2.5 mm to about 0.5 mm for the 1.38 wt % CMC solution. Considering that the confined-free film within the window was being fed from its surrounding walls, the liquid film thickness on the wall within the window region had been reduced by 90% compared with that on the upper wall. Interestingly, with the decrease of Re_{eff} , the film in the lower half of the window became thicker, and the position of the minimum film thickness gradually moved upstream. The film thickness was always symmetrically distributed with $X = 0$, as shown in Figure 2f. Similar conclusions have also been found in our previous research on the twin-liquid film of Newtonian fluid.³⁵ When the film approaches the lower wall ($Y = 48 \text{ mm}$), the thickness increases sharply to more than 2.5 mm. More about the hydrodynamics of twin-liquid films with non-Newtonian fluids will be discussed by the numerical simulations below.

3. NUMERICAL SIMULATIONS

For more details on hydrodynamics, the CMC solution film falling over a vertical plate with an opening window along the gravity direction was considered. The opening window has a length of L and a width of W . The computational domain and boundary conditions for the numerical simulation are shown in Figure 3a. More details were provided in our previous work.³⁵

3.1. Governing Equations and Boundary Conditions.

3.1.1. Hydrodynamics. The flow process is assumed to be isothermal. Air is a Newtonian fluid, and the viscosity of the CMC solution is set according to the rheological test results in Section 2.2. Both phases are incompressible. Since the density and viscosity of the CMC solution are higher than those of air by several orders of magnitude, the influence of air on the film flow is ignored.

The governing equations for film flow are as follows:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (14)$$

The equation of motion is

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \bar{\boldsymbol{\tau}} + \rho \mathbf{g} + F_{\text{vol}} \quad (15)$$

where $\bar{\boldsymbol{\tau}}$ is the shear stress, $\mathbf{g}$ represents gravitational acceleration, and F_{vol} is the additional volume force transformed from the continuum surface force (CSF) approach, existing only on the free surface.

$$F_{\text{vol}} = \sigma \frac{\rho \kappa \alpha_L}{(1/2)(\rho_L + \rho_G)} \quad (16)$$

in which σ is the surface tension coefficient and κ is the curvature of the interface, is defined as

$$\kappa = \nabla \cdot \frac{\mathbf{n}}{|\mathbf{n}|} = -\frac{1}{|\mathbf{n}|} \left[\frac{\mathbf{n}}{|\mathbf{n}|} \cdot \nabla |\mathbf{n}| - \nabla \cdot \mathbf{n} \right] \quad (17)$$

where $\mathbf{n}$ is the normal vector of the interface.

For non-Newtonian fluids, the shear stress can similarly be expressed in terms of a non-Newtonian viscosity μ , where μ is a function of the shear rate $\dot{\gamma}$:

$$\bar{\boldsymbol{\tau}} = \mu(\dot{\gamma}) \quad (18)$$

$\dot{\gamma}$ is a function of all three invariants of the rate-of-deformation tensor $\bar{\mathbf{D}}$:

$$\dot{\gamma} = \sqrt{\frac{1}{2} \bar{\mathbf{D}} : \bar{\mathbf{D}}} \quad (19)$$

$\bar{\mathbf{D}}$ is defined by

$$\bar{\mathbf{D}} = \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right) \quad (20)$$

The volume of fluid (VOF) model is adopted to capture the phase interface, which calculates the transport equation of the phase fractions. The physical properties of the mixed interface adopt a weighted average value.

$$\frac{\partial (\alpha_L \rho_L)}{\partial t} + \nabla \cdot (\alpha_L \rho_L \mathbf{u}_L) = 0 \quad (21)$$

$$\begin{aligned} \rho &= \alpha_L \rho_L + (1 - \alpha_L) \rho_G \\ \mu &= \alpha_L \mu_L + (1 - \alpha_L) \mu_G \\ \alpha_L + \alpha_G &= 1 \end{aligned} \quad (22)$$

where α is the phase fraction, μ and ρ are the dynamic viscosity and density of the fluids, respectively, and the subscripts L and G indicate the liquid and gas phases, respectively.

The laminar falling film analytical solution of a power-law non-Newtonian fluid is adopted to define the initial conditions for the solution. The film inlet velocity (U) follows the velocity distribution shown in eq 7, and the initial film thickness (δ_0) is consistent with the analytical solution of eq 8. The computational domain in the film thickness direction (z direction) is set to be 6 times the initial film thickness, taking into account both accuracy and computational cost. All boundary conditions are listed in Table 2.

3.1.2. Mass Transfer. The mass transfer performance of the twin-liquid film is investigated through the process of absorbing CO_2 by a highly viscous non-Newtonian liquid. In addition to

Table 2. Boundary Conditions

Region	Boundary type	Boundary conditions
Liquid inlet	Velocity inlet	$U = \left(\frac{\rho g \delta_0}{K} \right)^{1/n} \frac{n \delta_0}{n+1} \left[1 - \left(\frac{y}{\delta_0} \right)^{n+1/n} \right]$, $\alpha_L = 1$, $v = w = 0$
Gas inlet	Pressure inlet	$p = 0$ (gauge pressure), $\alpha_L = 1$
Liquid/gas outlet	Pressure outlet	$p = 0$ (gauge pressure)
Wall	Wall	$u = v = w = 0$
Window	Symmetric	$\frac{\partial u}{\partial z} = 0$, $\frac{\partial v}{\partial z} = 0$, $\frac{\partial w}{\partial z} = 0$, $\frac{\partial \alpha_L}{\partial z} = 0$
Far field	Symmetric	$\frac{\partial u}{\partial z} = 0$, $\frac{\partial v}{\partial z} = 0$, $\frac{\partial w}{\partial z} = 0$, $\frac{\partial \alpha_L}{\partial z} = 0$

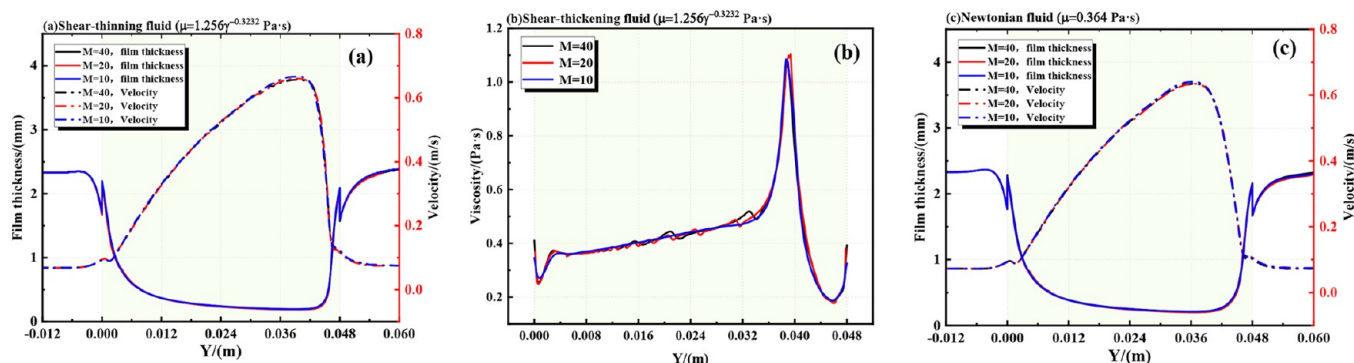


Figure 4. Grid independence verification. (a) shear-thinning fluid; (b) shear-thickening fluid; and (c) Newtonian fluid. ($\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, $\text{Re}_{\text{eff}} = 0.311$, $W = 24 \text{ mm}$, and $L = 48 \text{ mm}$).

the hydrodynamic assumptions stated above, the absorption is assumed to be a convective mass transfer process. The concentration of CO_2 in the liquid is very low, and the species is assumed to be physicochemically inert (no reaction and no surface activity). The absorbent's physical properties (i.e., density and viscosity) are considered constant during the absorption process. In addition, since the diffusivity of CO_2 in the gas is several orders of magnitude higher than in the liquid, the gas and liquid phases are considered to reach an equilibrium state at the gas–liquid interface.³⁶ The equilibrium concentration of CO_2 at the interface follows Henry's law.

The mass transport equation for CO_2 gas absorption in the liquid phase is

$$\nabla(\rho u C_{\text{CO}_2}) - \nabla \cdot \rho D_{\text{CO}_2} \nabla C_{\text{CO}_2} = 0 \quad (23)$$

where C_{CO_2} and D_{CO_2} are, respectively, the mass fraction in the liquid and the diffusion coefficient of CO_2 .

At the gas–liquid interface, the concentration of CO_2 reaches the equilibrium concentration $C_{\text{CO}_2}^*$, and the mass fraction of CO_2 at the gas–liquid interface is

$$C_{\text{CO}_2} = C_{\text{CO}_2}^* \quad (24)$$

In addition, the momentum boundary condition at the interface is set to the slip wall.

The boundary condition at the inlet is

$$C_{\text{CO}_2} = 0 \quad (25)$$

At the other boundaries, the homogeneous Neumann boundary condition is chosen as

$$\frac{\partial C_{\text{CO}_2}}{\partial \mathbf{n}} = 0 \quad (26)$$

where $\mathbf{n}$ is the face normal.

3.2. Numerical Solution Strategy. The simulation strategy diagram is shown in Figure 3b. In Step 1, obtain the position coordinates of the gas–liquid interface through the VOF method. Due to the stable film flow with highly viscous fluid, its gas–liquid interface undergoes almost no change with time. Then, in Step 2, geometric software is used to extract the interface position coordinates and merge this interface with only the liquid bulk, excluding the gas bulk. Hereafter, the computational domain is reconstructed. Set the gas–liquid interface as a free sliding wall boundary (i.e., no resistance to flow) and solve the continuity equation and momentum

equation for the liquid single phase. In Step 3, after the flow stabilizes, solve the convection–diffusion equation. Similar methods have been adopted by Ishikawa et al.,³⁷ Wasden et al.,¹⁴ and Chasanis et al.³⁸

In Step 1, the VOF model is adopted to obtain the position of the free surface, and the pressure implicit splitting (PISO) algorithm is used for the iterative solution of pressure–velocity coupling. The second-order upwind style is adopted for convection terms, and the PRESTO format for pressure. The second-order diffusion term and velocity gradient utilize the gradient algorithm based on the Green–Gauss Node to ensure second-order spatial accuracy. For interface tracking, a geometric reconstruction scheme (Geo-Reconstruct) based on the PLIC format is used. The discretization of the nonsteady term is second-order implicit. To ensure the efficiency and accuracy of the calculation, an adaptive time step is adopted, and the Courant number is always kept less than 0.25 during the simulation. For Step 2, the species transfer model is adopted to solve the concentration distribution for the reconstructed computational domain. A second-order upwind scheme is chosen for the convective terms, and a central difference scheme is selected for the diffusive terms. For pressure–velocity coupling, the SIMPLE scheme is used for the iterative solution. The discretization of the momentum and the species transport equation employs the second-order up-wind scheme.

3.3. Grid Independence Check and Verification.

3.3.1. Hydrodynamics. The computational domain is discretized in Cartesian space using structured grids, and the corresponding nondimensional grid size Δ_X , Δ_Y , and Δ_Z are defined as follows:

$$\begin{aligned} \Delta_X &= \Delta_Y = \frac{R}{M} \\ \Delta_Z &= \frac{1}{M} \end{aligned} \quad (27)$$

Where M represents the number of grids in free film thickness, and R represents the ratio of grid height (Δ_X) and length (Δ_Y). Since the film flow variation in the X and Y directions is less sensitive than that in the Z direction, and the high-viscous film has no intensive surface wave, we set $R = 4$, identical to our previous work. A comparison of the film thickness, velocity, and viscosity distribution with different mesh densities is shown in Figure 4. It can be observed that the simulation results for $M = 10$, $M = 20$, and $M = 30$ are very close, with the relative deviation being less than 1%, which ensures grid-independent simulation results. Considering computational cost and accuracy, $M = 20$ is

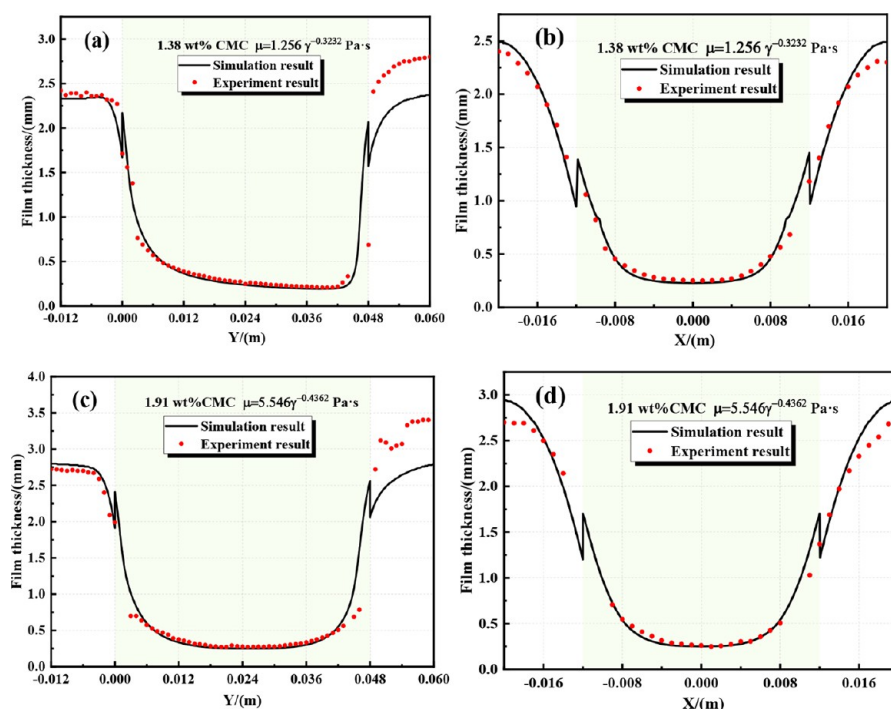


Figure 5. Comparison of the film thickness distribution at $X = 0$ (a, c) and $Y = 0.024$ mm (b, d) by numerical simulation and experiment results ($W = 24$ mm, $L = 48$ mm).

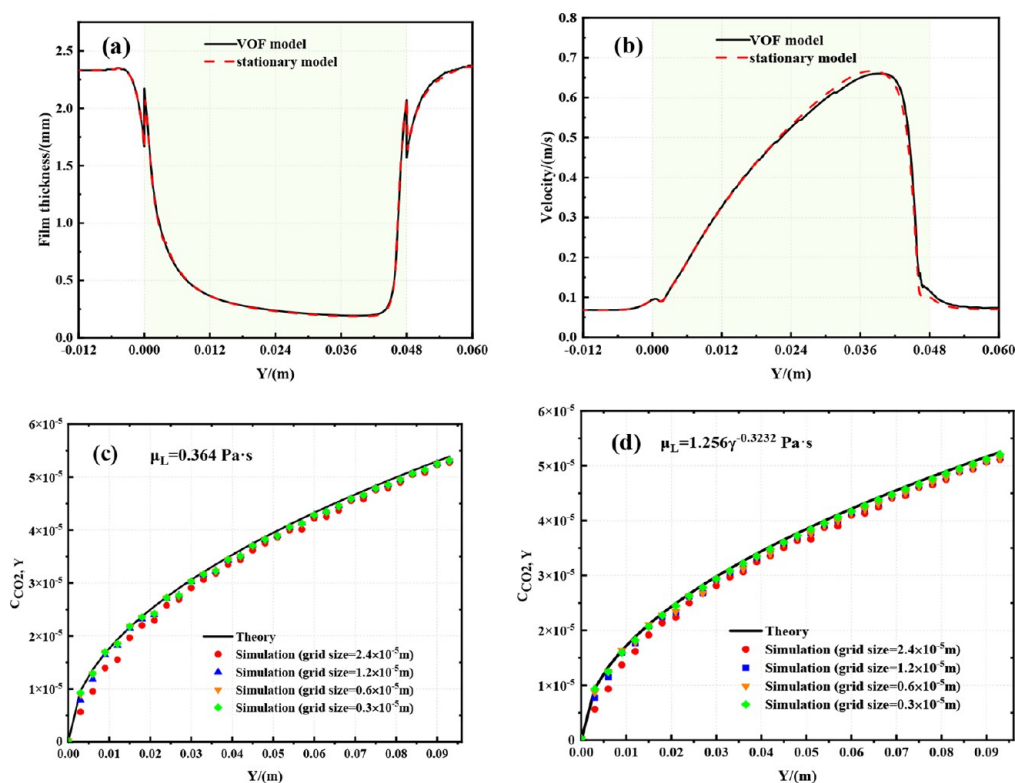


Figure 6. Film thickness (a) and velocity (b) calculated by VOF and Stationary model; the comparison of CO_2 mass fraction along the flow direction between simulated and theoretical values with different grid density for Newtonian fluid (c) and 1.38 wt % CMC aqueous solution (d). The theoretical values in (c) and (d) are derived from the Bird's work.²⁰

adopted in the present work. Figure 5 compares the film thickness results with the two CMC solutions in Figure 1 obtained through numerical simulation and experimental work. It can be observed that the simulated film thickness agrees well

with the experimental results in general. The difference at the lower wall ($Y > 0.048$ m) is due to the slight tilt of the vertical plate during experiments, which caused more liquid to flow

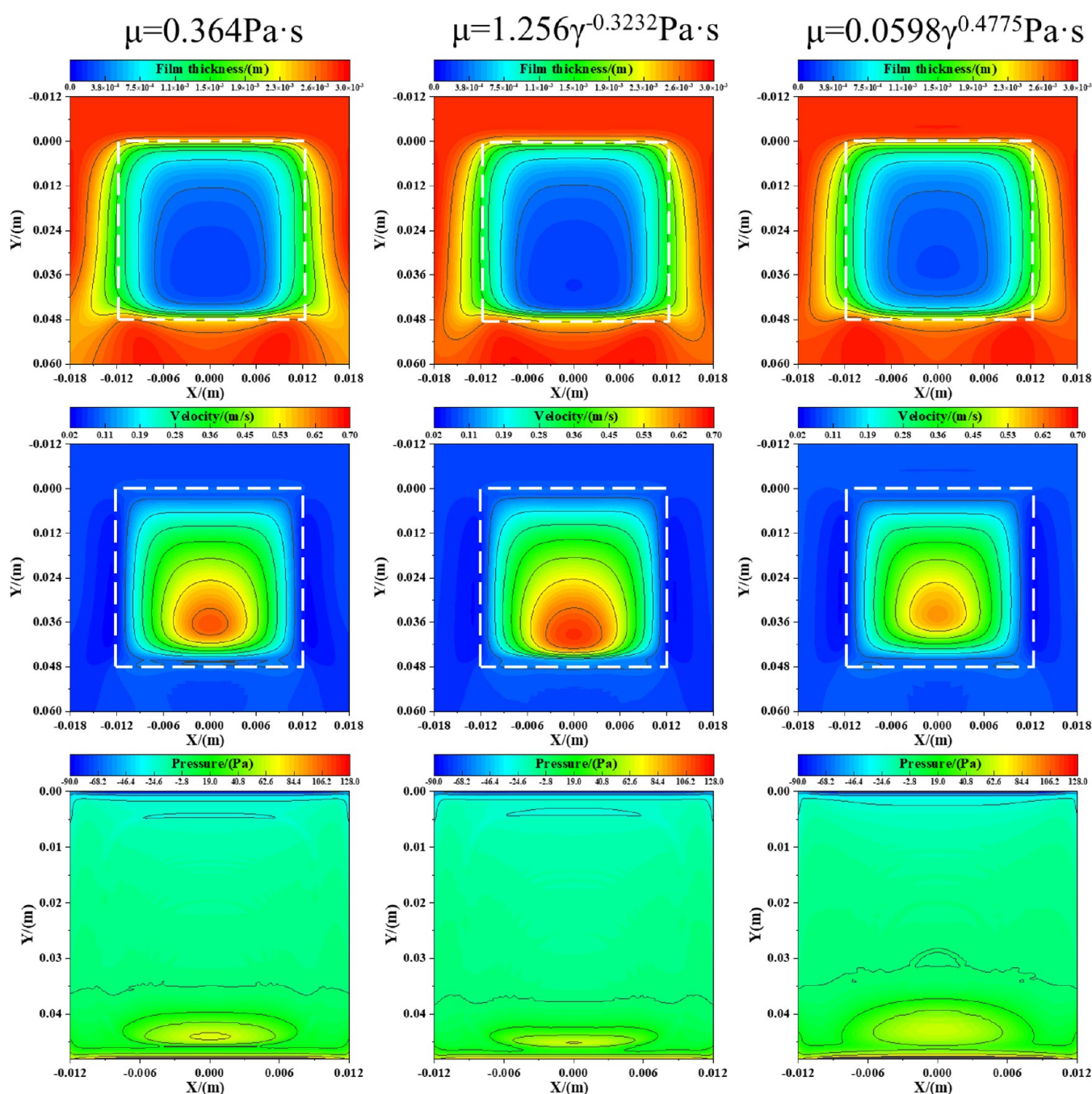


Figure 7. Film thickness, surface velocity, and pressure distribution of the twin-liquid film with different rheological properties. Window region is marked with white dashed line ($Re_{eff} = 0.311$, $\mu_{eff} = 0.364$ Pa·s, $W = 24$ mm, and $L = 48$ mm).

toward the front side of the plate, making the liquid film on the lower wall thicker than the simulation result.

3.3.2. Mass Transfer. Figure 6a,b shows the film thickness and film velocity calculated by the VOF and Stationary models for the vertical plate with an opening window. The window's length is 0.048 m, which corresponds to $Y \in (0, 0.048)$. It can be observed that the results of film location and velocity from both models are almost consistent, which confirms the effectiveness of the film flow field by the Stationary model.

According to the theory of solute penetration, the gas–liquid contact time in the present absorption process is short, and the solute cannot penetrate deeper into the liquid film; therefore, the solute permeation model is applicable. According to the

work of Pigford and Johnstone,²⁶ the theoretical amount of CO_2 absorbed (B_{CO_2}) by a flat Nusselt film of length Y and width X is

$$B_{CO_2,Y} = XYC_{CO_2}^* \sqrt{\frac{4D_{CO_2}U_0}{\pi Y}} \quad (28)$$

where $C_{CO_2}^*$ is the CO_2 equilibrium concentration at the gas–liquid interface, D_{CO_2} is the diffusion coefficient of CO_2 , and U_0 is the velocity at the interface. Thus, the mass fraction of CO_2 over the cross-section at Y can be defined as the ratio of the CO_2 mass flow rate ($B_{CO_2,Y}^*$) to the liquid mass flow rate (B_{liquid}):

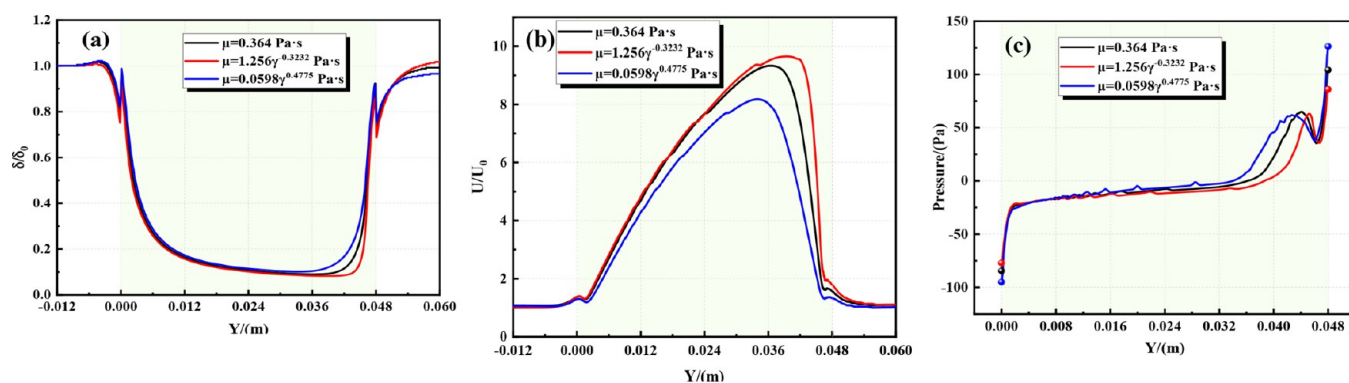


Figure 8. Film thickness (a), surface velocity (b), and pressure (c) distribution of the twin-liquid film with different rheological properties at $X = 0$. The solid sphere in the figure represents the pressure value at the window frame ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, $W = 24 \text{ mm}$, and $L = 48 \text{ mm}$).

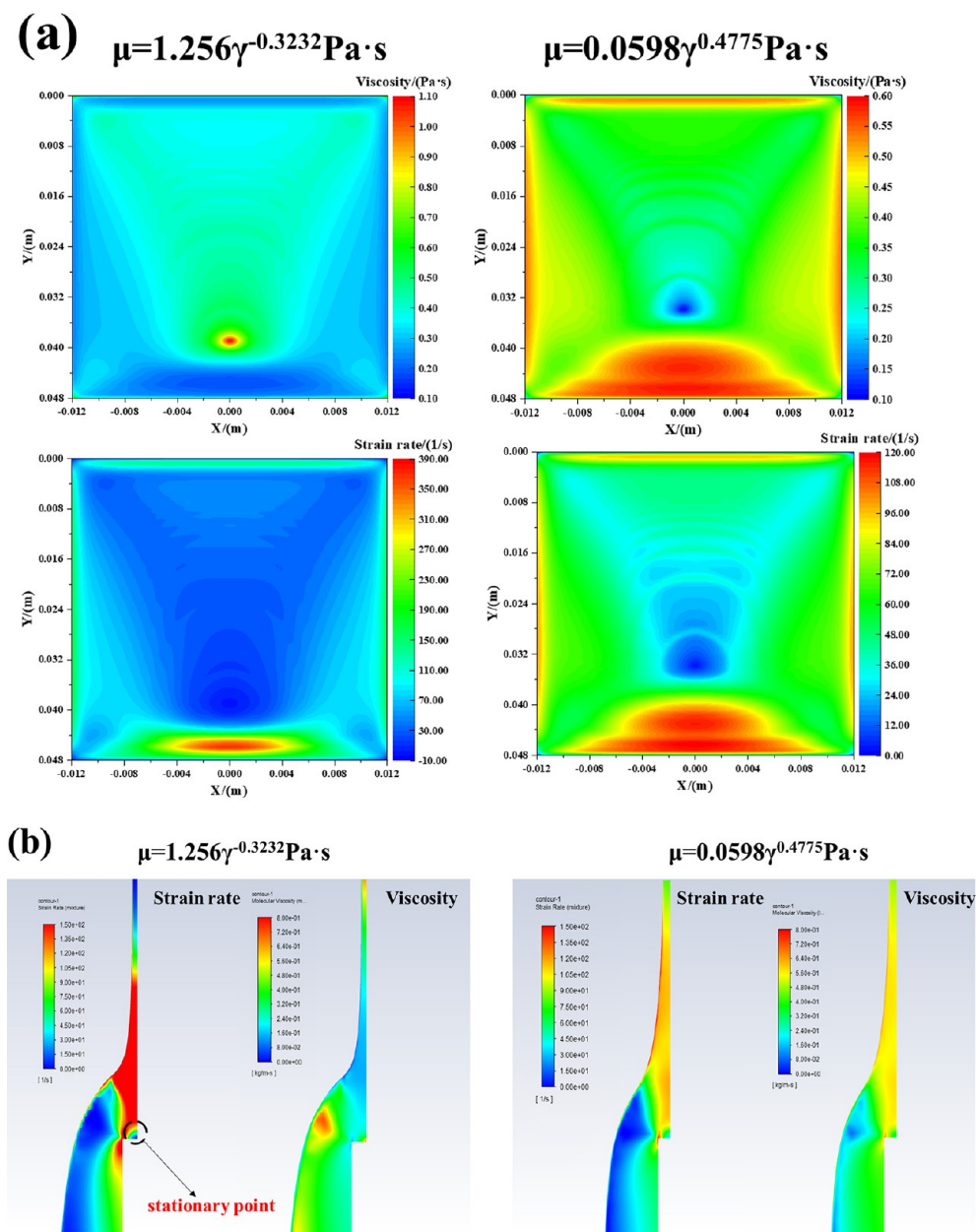


Figure 9. Viscosity and strain rate distribution with different rheological properties. (a) XY plane and (b) YZ plane ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, $W = 24 \text{ mm}$, and $L = 48 \text{ mm}$).

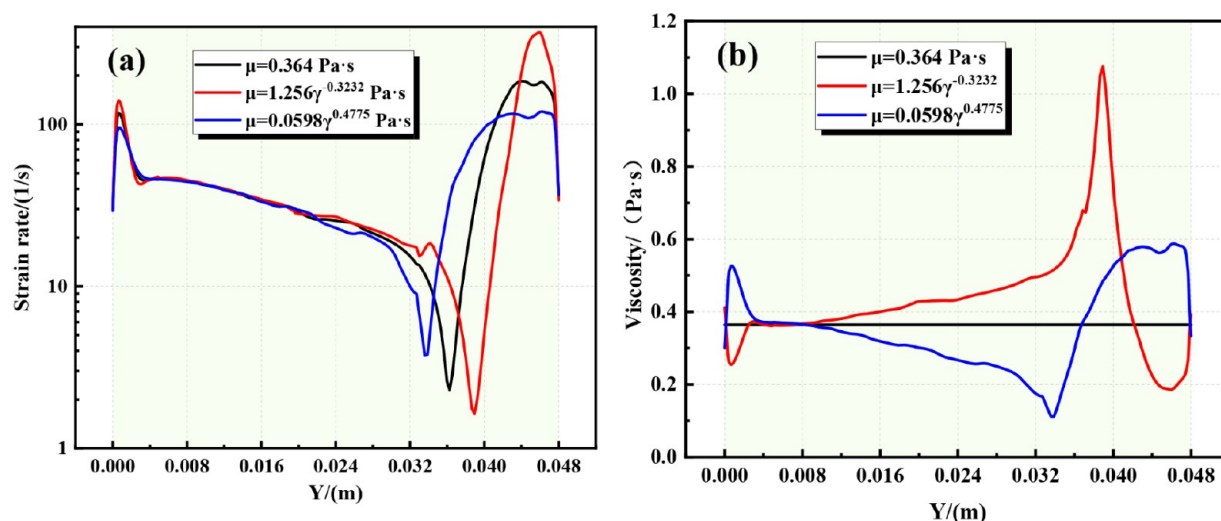


Figure 10. Strain rate (a) and viscosity (b) distribution with different rheological properties at $X = 0$ ($Re_{eff} = 0.311$, $\mu_{eff} = 0.364$ Pa·s, $W = 24$ mm, and $L = 48$ mm).

$$C_{CO_2,Y} = \frac{B_{CO_2,Y}}{B_{liquid}} = \frac{XYC_{CO_2}^* \sqrt{\frac{4D_{CO_2}U_0}{\pi Y}}}{XQ} = \frac{YC_{CO_2}^* \sqrt{\frac{4D_{CO_2}U_0}{\pi Y}}}{Q} \quad (29)$$

For a Newtonian fluid,

$$U_{max} = 1.5\bar{U} = 1.5 \left(\frac{Re^2 \mu g}{3\rho_L} \right)^{1/3} \quad (30)$$

For a non-Newtonian fluid,

$$U_0 = \left(\frac{\rho g \delta^n}{K} \right)^{1/n} \frac{n\delta}{n+1} \quad (31)$$

To verify the accuracy of the simulation results, the mass transfer process of the falling liquid film with a solid vertical plate has been simulated. The length of the solid plate is 0.096 m, and its width is 0.04 m. For the Newtonian fluid, its viscosity is $\mu_L = 0.364$ Pa·s, and $\mu_L = 1.256\gamma^{-0.3232}$ Pa·s for the non-Newtonian fluid. All of the volume flow rates per unit plate width are $Q = 1.13 \times 10^{-4}$ m²/s, $\rho_L = 1000$ kg/m³, and $Re_{eff} = 0.31$ for both the Newtonian and non-Newtonian fluids. The equilibrium concentration ($C_{CO_2}^*$) is set to 1.716×10^{-3} and the diffusion coefficient D_{CO_2} is set as 1.73×10^{-9} m²/s during the verification of grid independence, which is based on the diffusion of CO₂ in pure water. The comparison results of the CO₂ mass fraction along the flow direction (Y direction) between the simulated value and theoretical value with different grid densities are shown in Figure 6c,d. When the grid size of the concentration boundary layer is reduced to 0.3×10^{-5} m, the difference between the simulation results and the analytical results from eq 31 is less than 3%. Hence, it can be considered that the simulation results agree well with the theoretical value, and grid independence has been achieved with this grid size.

4. RESULT AND DISCUSSION

4.1. Flow Pattern of the Twin-Liquid Film with Non-Newtonian Fluid. The hydrodynamics of the twin-liquid films with different rheological properties but the same Re_{eff} and μ_{eff} values have been investigated. As shown in Figure 7, the flow

fields with different rheological properties are generally similar. The confined-free film thickness is reduced by 90%, and its velocity is increased by 8–10 times compared with that of the wall-bounded film on the upper wall ($Y < 0$). The highest and lowest static pressures are located near the downstream and upstream window frames, respectively.

However, the values of film thickness and surface velocity of the confined-free film with different rheological properties are not consistent. As revealed in Figure 8, δ_0 and U_0 represent the theoretical values of film thickness and velocity obtained from eqs 7 and 8, respectively. The maximum surface velocity for the confined-free film with the Newtonian fluid is 10% lower than that with shear-thinning fluid ($n < 1$) and 10% higher than that with shear-thickening fluid ($n > 1$) when approaching the downstream window frame. Correspondingly, the confined-free film thickness exhibits an opposite trend due to mass conservation ($\bar{U}\delta = \text{constant}$). But on the lower window frame ($Y > 0.048$ m), the thickness of the wall-bounded film with shear-thinning fluid is the greatest. In addition, both the positions for the minimum film thickness and the highest velocity shift toward the downstream window frame as the fluid transitions from shear-thickening to shear-thinning.

Moreover, the liquid's passage across the window frame resembles the two-dimensional step flow studied by Bontozoglou and Serifi,³⁹ with low pressure near the upstream frame causing the liquid's lateral deflection into the window and high pressure at the downstream frame pushing the liquid back to the wall. This pressure gradient between the upstream and downstream frames results in confined-free film deceleration and thickening. Fluids with different rheological properties exhibit different pressure values at the window frame ($Y = 0.048$ m), as shown in Figure 8c. The pressures for the film are 104 Pa for the Newtonian fluid, 86 Pa for the shear-thinning fluid, and 126 Pa for the shear-thickening fluid. Hence, this greater adverse pressure gradient for the shear-thickening fluid results in lower velocity and higher film thickness, as shown in Figure 8a,b.

Despite these fluids having the same Re_{eff} , the specific rheological properties of the fluid determine their unique film dynamics. The opening window on the solid plate introduces free film flow instead of wall-bound film flow exclusively. The main total forces, including wall shear stress, viscous force, and gravity force, are applied to the wall-bound film flow. While,

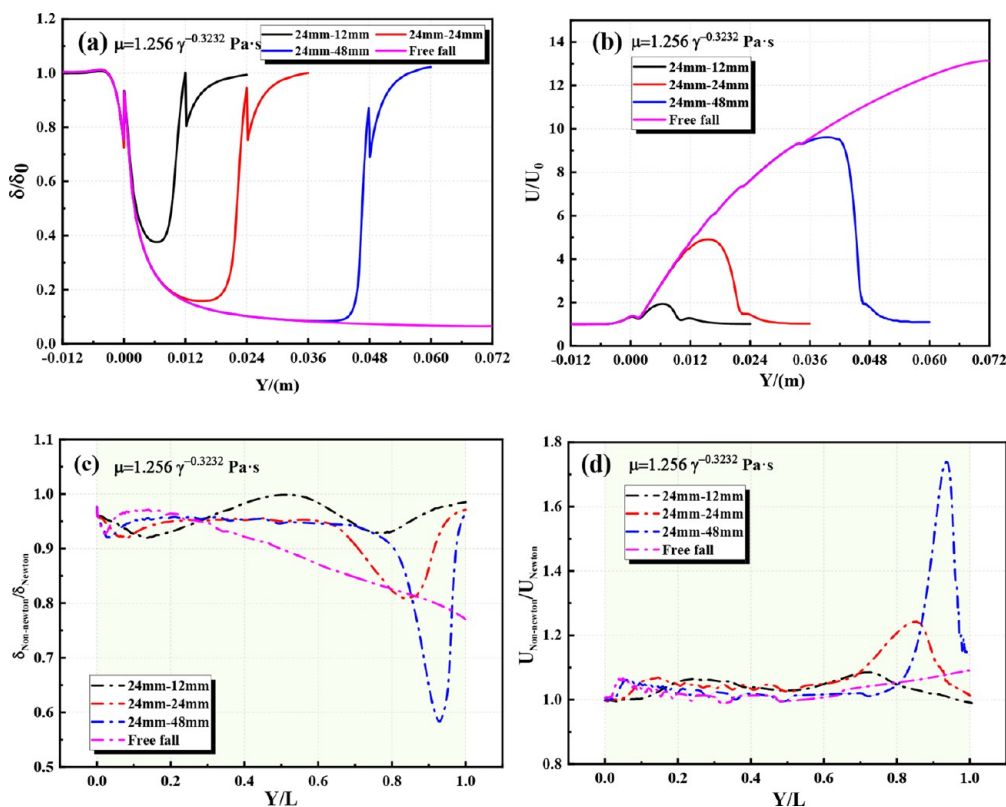


Figure 11. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$ distribution for shear-thinning fluid with different window length at $X = 0$ ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, and $W = 24 \text{ mm}$).

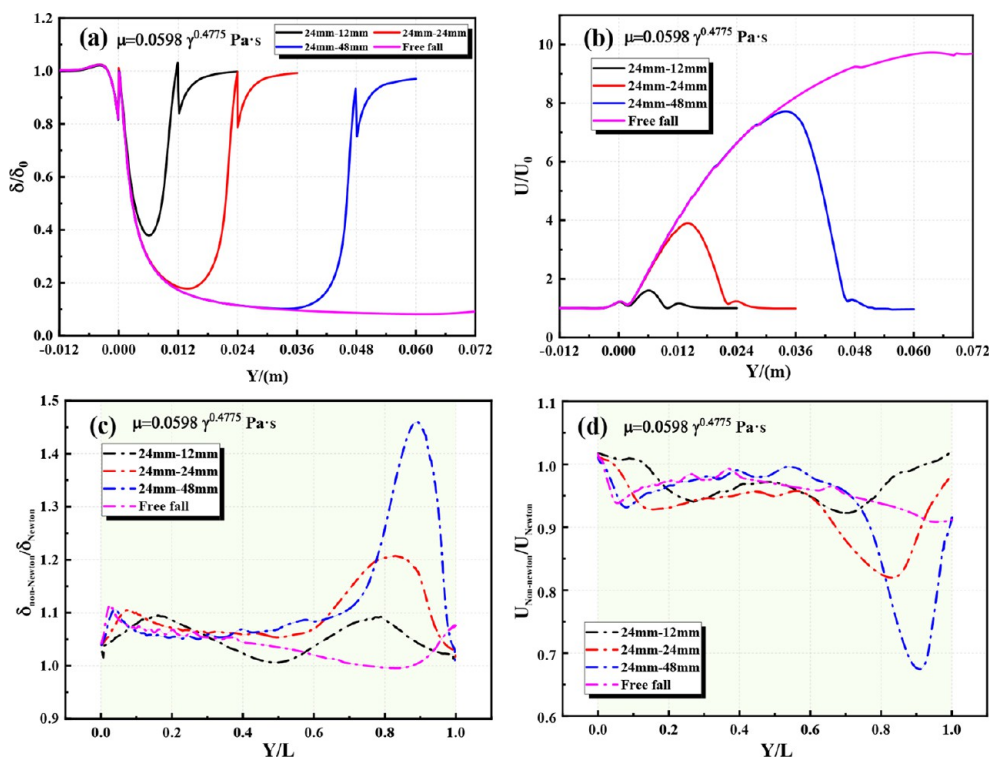


Figure 12. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$ distribution for shear-thickening fluid with different window length at $X = 0$ ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, and $W = 24 \text{ mm}$).

within the window, wall shear stress disappears, while viscous force and gravity force remain active.^{2,28} Since viscosity is related to strain rate, different film characteristics with non-

Newtonian fluids are produced compared to those with Newtonian fluid, as shown in Figures 9 and 10. The viscosity of the shear-thinning fluid ($\mu = 1.256\gamma^{-0.3232}$) and shear-

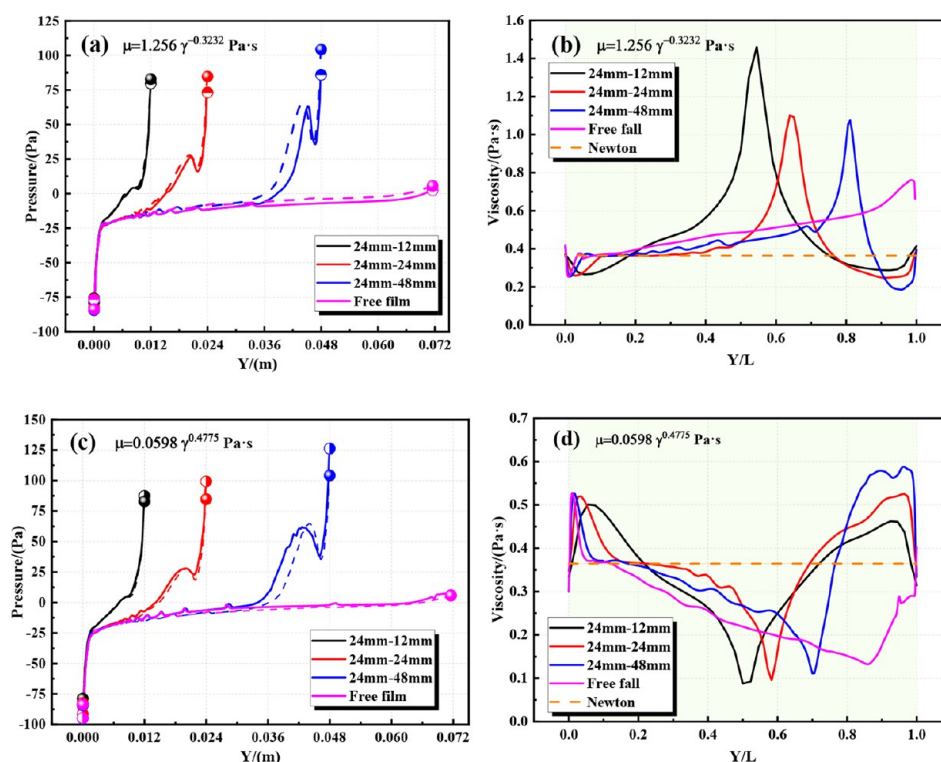


Figure 13. Pressure and viscosity distribution for shear-thinning fluid (a, b) and shear-thickening fluid (c, d) with different window length at YZ plane. Solid line: non-Newtonian fluid; dashed line: Newtonian fluid. The sphere represents the pressure value at the window frame. ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364$ Pa·s, and $W = 24$ mm).

thickening fluid ($\mu = 0.0598\gamma^{0.4775}$) is symmetrical along $X = 0$ within the window. At the center of the window, since gravity and the adverse pressure gradient generated between the downstream and upstream window frames offset each other, the velocity gradient gradually decreases, resulting in a low strain rate. Near the window frames, the sudden transition of boundary conditions causes the film velocity to change dramatically, leading to strong shear stress. Particularly near the downstream window frame, the film velocity decreases rapidly within a short distance, achieving the largest strain rate. For shear-thinning fluids, the viscosity is negatively correlated with the shear rate. With the development of the confined-free film, its velocity gradually reaches a maximum, and the strain rate is reduced accordingly. Hence, its viscosity is gradually increases along the flow direction and reaches a maximum of 1.1 Pa·s at $Y = 0.038$ mm. The maximum local viscosity is about 3 times the effective viscosity. Subsequently, a stagnation flow is formed at the downstream window frame, as shown in Figure 9b, causing the strain rate to increase and the viscosity to decrease. On the contrary, due to the viscosity of shear-thickening fluid being positively related to strain rate, the distribution of viscosity along the flow direction is opposite to that of shear-thinning fluid, as shown in Figure 10b. Moreover, as revealed by eq 15 lower shear stress leads to lower static pressure. Comparatively, the shear-thinning behavior is more conducive to the acceleration and thinning of confined-free films due to the lower viscosity near the downstream window frame, while the shear-thickening behavior hinders this trend, as shown in Figure 8a,b.

4.2. Hydrodynamics. 4.2.1. Effect of Opening Window Geometry. (1) Window Length. According to our previous work on Newtonian twin-liquid films,³⁵ window width primarily provides a large area for liquid spatial distribution to reduce film

thickness, while window length plays a major role in film acceleration and thinning, which is relatively crucial for the hydrodynamics and mass transfer performance of the twin-liquid film. This section focuses on the influence of the window length.

For non-Newtonian fluids, the increase in window length is also beneficial to the thinning and acceleration of the confined-free film. As the window length extends from 12 mm to 48 mm, the film's minimum thickness decreases from a quarter to a tenth of the Nusselt theoretical value $\delta_0 = (\frac{3\mu_{\text{eff}}Q}{\rho_L g})^{1/3}$, while the maximum velocity surges from twice to 10-fold the Nusselt theoretical interface velocity $U_0 = 1.5(\frac{Re_{\text{eff}}^2 \mu_{\text{eff}} g}{3\rho_L})^{1/3}$, as depicted in

Figures 11 and 12. Initially, the twin-liquid film thickness and velocity are consistent with those of the free-falling film, which develops freely without the downstream window frame. With the twin-liquid film development, it gradually deviates from the free-falling film due to the influence of the high-pressure region at the downstream window frame and the viscous force from the side frames. For the free-falling film, the difference in film thickness and velocity between Newtonian fluid and non-Newtonian fluid is less than 10% under the same Re_{eff} and μ_{eff} . But for the twin-liquid film, the difference in film thickness and velocity near the downstream window frame between non-Newtonian fluid and Newtonian fluid increases with the window length, as shown in Figures 11c,d and 12c,d. This indicates that the difference in the twin-liquid film's thickness and velocity between Newtonian fluid and non-Newtonian fluid with the same Re_{eff} and μ_{eff} is mainly attributed to the presence of the high-pressure zone near the downstream window frame.

Moreover, a longer window length boosts the confined-free film's acceleration distance, resulting in higher velocity. As a

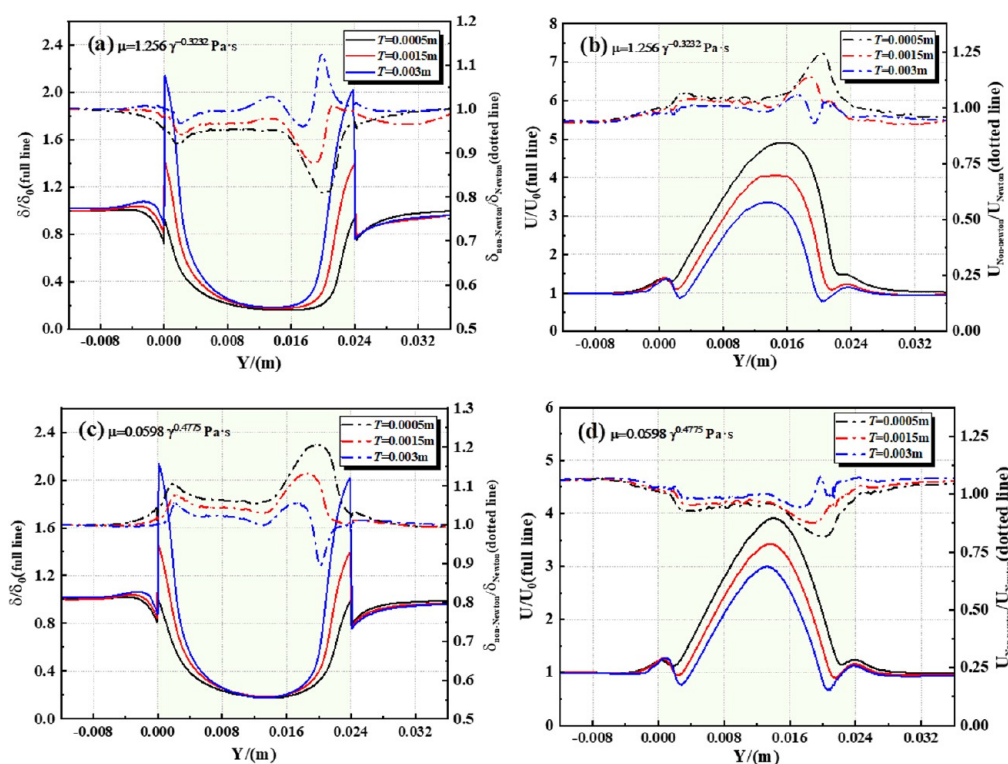


Figure 14. Solid line: dimensionless film thickness and dimensionless velocity distribution for shear-thinning fluid (a, b) and shear-thickening fluid (c, d) with different plate thickness at $X = 0$. Dashed line: $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and $U_{\text{Non-newton}}/U_{\text{Newton}}$ distribution for shear-thinning fluid (a, b) and shear-thickening fluid (c, d) with different wall thickness at $X = 0$ ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

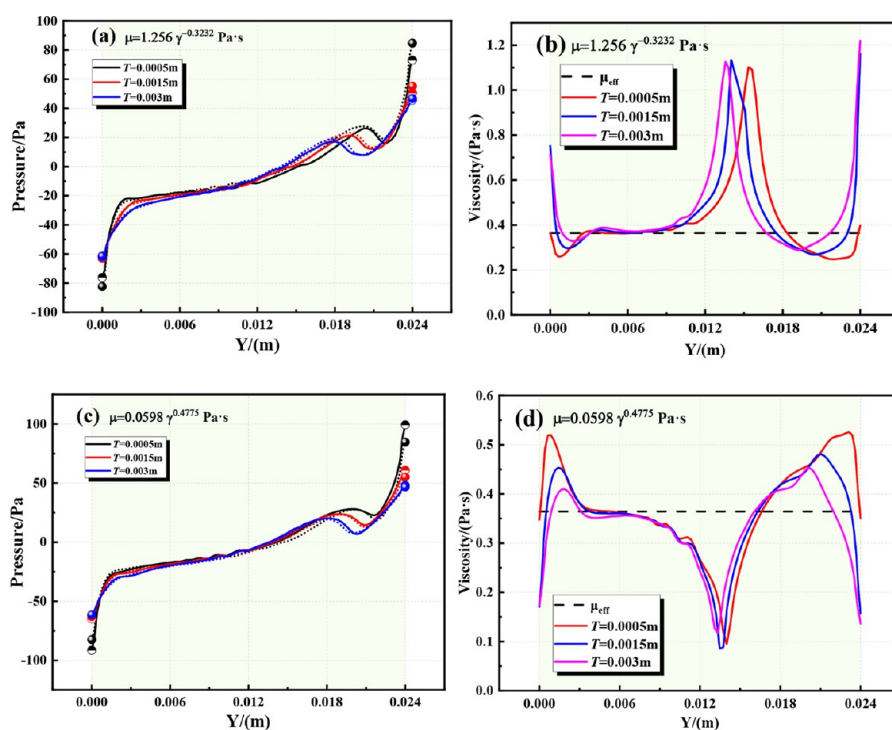


Figure 15. Pressure and viscosity distribution for shear-thinning fluid (a, b) and shear-thickening fluid (c, d) with different wall thickness at $X = 0$. Solid line: non-Newtonian fluid; dashed line: Newtonian fluid. The sphere represents the pressure value at the window frame ($Re_{\text{eff}} = 0.311$, $\mu_{\text{eff}} = 0.364 \text{ Pa}\cdot\text{s}$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

result, the film velocity changes more violently near the stagnation point in the downstream window frame, which generates a higher strain rate. According to eq 15, the pressure P is directly proportional to the shear stress, so the difference in

pressure between the shear-thinning fluid and Newtonian fluid at the downstream window frame increases with the window length, as shown in Figure 14a. The shear-thickening fluids exhibit the same trend, as shown in Figure 13c.

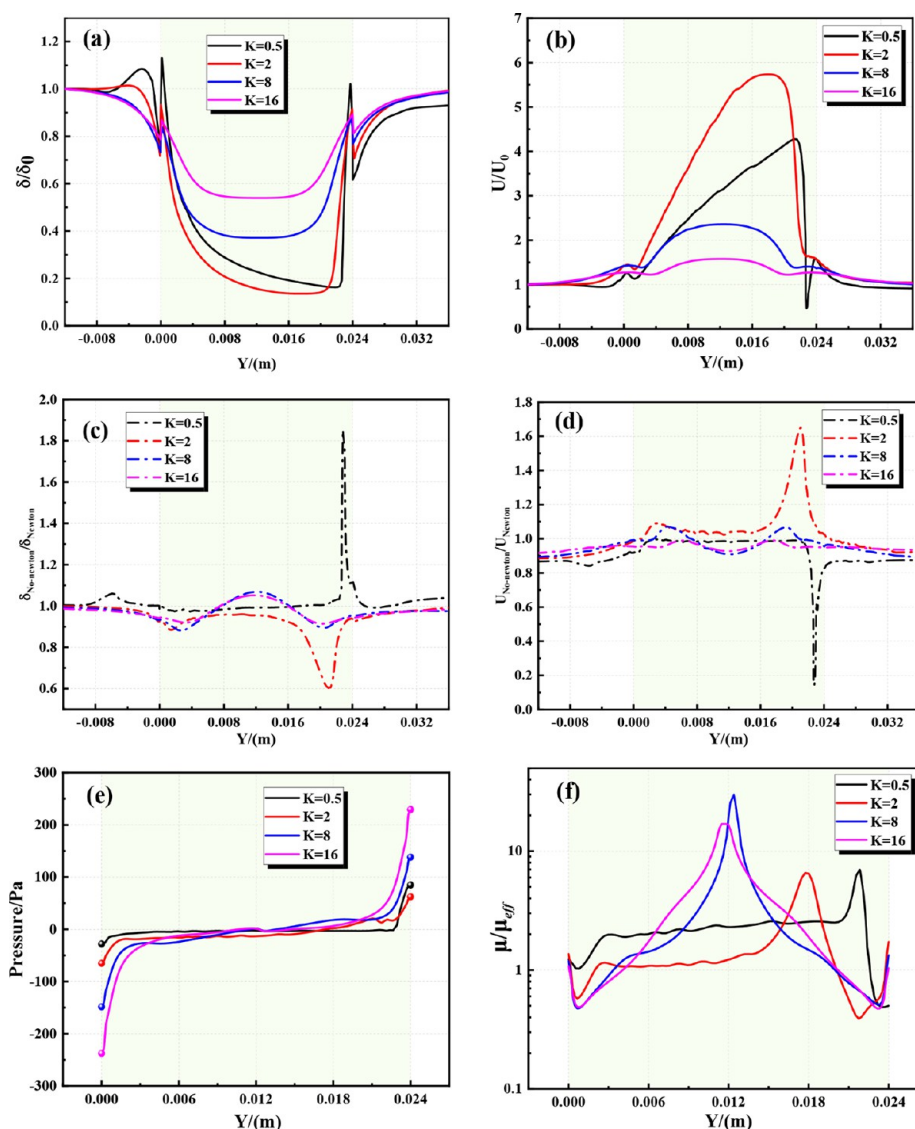


Figure 16. Hydrodynamics of shear-thinning fluid fluid with different consistency coefficients (K) at $X = 0$. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$; (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$; (e) pressure; and (f) viscosity ($Q = 1.133 \times 10^{-4} \text{ m}^2/\text{s}$, $n = 0.5$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

(2) *Plate Thickness.* The liquid crosses the window frame and then undergoes a transition between the wall-bound film and the confined-free film. The effects of plate thickness on the twin-liquid film have been investigated, as shown in Figure 14. With the increase in plate thickness, a higher liquid ridge forms at the upstream window frame ($-0.04 \text{ m} < Y < 0$) to generate a higher positive capillary pressure, which drives the liquid flow into the window. After the wall-bound film leaves the upstream window frame, the film velocity gradually changes from a parabolic distribution to a uniform distribution due to the action of fluid internal friction.² With the increase in plate thickness, the transition process from the wall-bound film to the confined-free film develops over a longer distance along the flow direction. On the other hand, the fluid internal friction counteracts part of the gravity, which reduces the film velocity near the upstream window frame. The same is true near the downstream window frame, where the film velocity of the thicker plate is lower. The lower initial velocity and shorter acceleration distance with a thicker plate ($T = 0.003 \text{ m}$) result in a 30% decrease in the film's maximum velocity in the window compared to that with a

thinner plate ($T = 0.0005 \text{ m}$). Meanwhile, the difference in film thickness and velocity between Newtonian and non-Newtonian fluids decreases from about 15% to 5% as the plate thickness increases from $T = 0.0005 \text{ m}$ to $T = 0.003 \text{ m}$, as shown in Figure 14. This is because the lower acceleration of the confined-free film leads to a lower strain rate, which suppresses the non-Newtonian effect. Hence, the local viscosity of the non-Newtonian fluid near the window frame becomes closer to the generalized effective viscosity (μ_{eff}) with the increase in plate thickness, as shown in Figure 15. The difference between the Newtonian fluid and non-Newtonian fluid decreases with the increase in plate thickness and this difference is mainly concentrated near the downstream window frame, as shown in Figure 14 (the dotted line represents the ratio of liquid film thickness and velocity between Newtonian and non-Newtonian fluids).

4.2.2. Effect of Rheological Properties. (1) *Consistency Coefficient.* The consistency coefficient K can be regarded as a measure of non-Newtonian fluid resistance at a specific shear rate. Figures 16 and 17 show the hydrodynamics of shear-

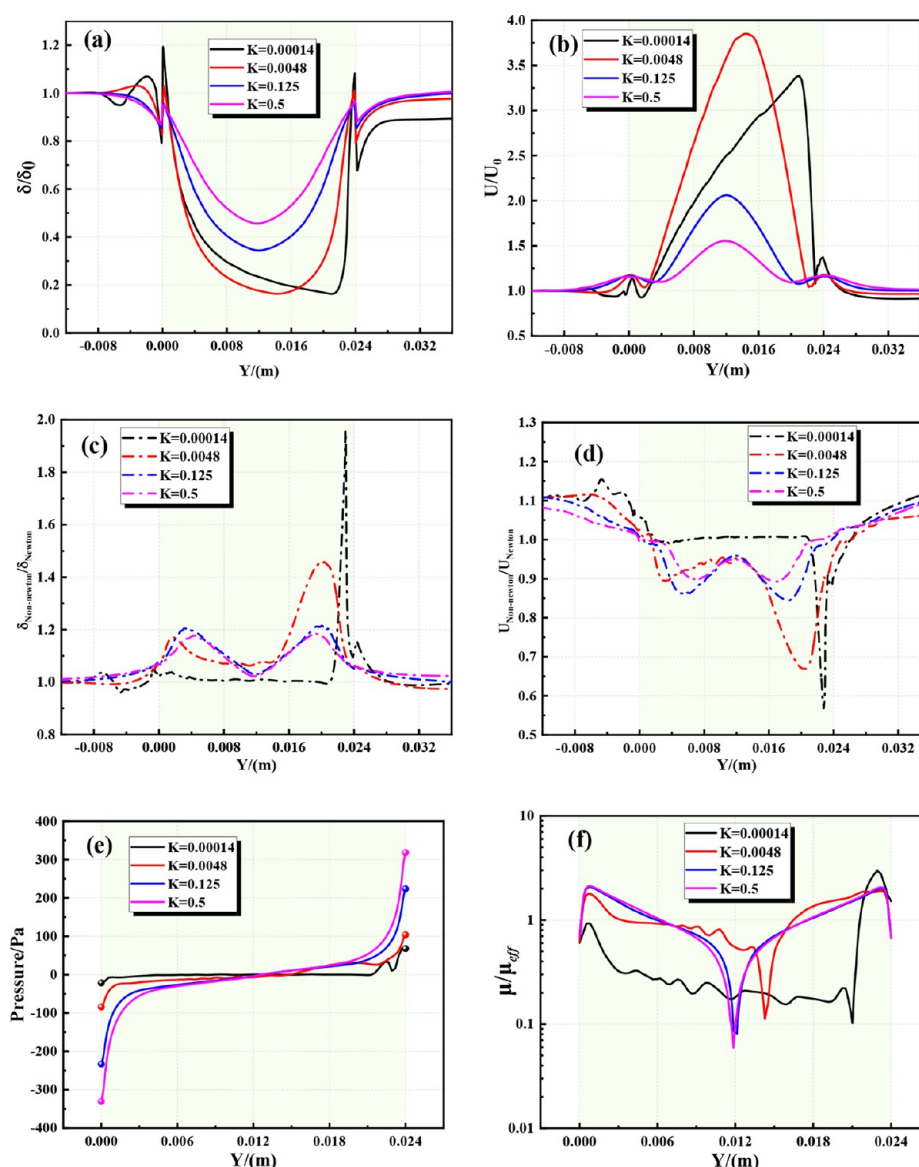


Figure 17. Hydrodynamics of shear-thickening fluid with different consistency coefficients (K) at $X = 0$. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$; (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$; (e) pressure; and (f) viscosity ($Q = 1.133 \times 10^{-4} \text{ m}^3/\text{s}$, $n = 2$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

thinning and shear-thickening twin-liquid films with different consistency coefficients, respectively. With the increase in K , the thinning and acceleration behavior of the confined-free films first increases and then decreases, which can be observed from the values of δ/δ_0 and U/U_0 , as shown in Figure 16a,b and Figure 17a,b. This trend can be well explained by the μ_{eff} and Re_{eff} . With the increase of K , μ_{eff} rises from tens of centipoises to thousands of centipoises. Correspondingly, the magnitude of Re_{eff} decreases from 10 to 1×10^{-2} . When Re_{eff} is greater than 1 (e.g., $K = 0.5$, $n = 0.5$ for shear-thinning and $K = 0.00014$, $n = 2$ for shear-thickening), the confined-free film is dominated by inertial force. At this point, the film is sufficiently thinned and accelerated within the window. When Re_{eff} is less than 1, the film is dominated by viscous force, and the thinning and acceleration effect of the liquid film is weakened.² Hence, a longer window is needed to accommodate higher μ_{eff} and achieve stronger thinning and acceleration effects of the confined-free film, as discussed in Section 4.2.1.

Although the influence of the consistency coefficient on Newtonian and non-Newtonian twin-liquid films can be globally unified through μ_{eff} and Re_{eff} , the local influence of non-Newtonian characteristics on twin-liquid films should not be ignored. The dimensionless confined-free film thickness $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and velocity $U_{\text{Non-newton}}/U_{\text{Newton}}$, which have the same μ_{eff} and Re_{eff} can partially describe the impact of non-Newtonian properties, as shown in Figures 16c,d and 17c,d. For the non-Newtonian film dominated by inertial forces, e.g., $K = 0.00014$, $n = 2$ in Figures 16d and 17d, a velocity stagnation point is generated near downstream window frames compared with the Newtonian fluid. This is because when the liquid film collides with the downstream window frame, it changes the shear stress distribution inside the film, leading to a rearrangement of local viscosity under the effects of shear-thinning or shear-thickening. This may have a profound impact on the stability of the non-Newtonian confined-free film. For the non-Newtonian film dominated by viscous forces, both the values of $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and $U_{\text{Non-newton}}/U_{\text{Newton}}$ tend to approach 1

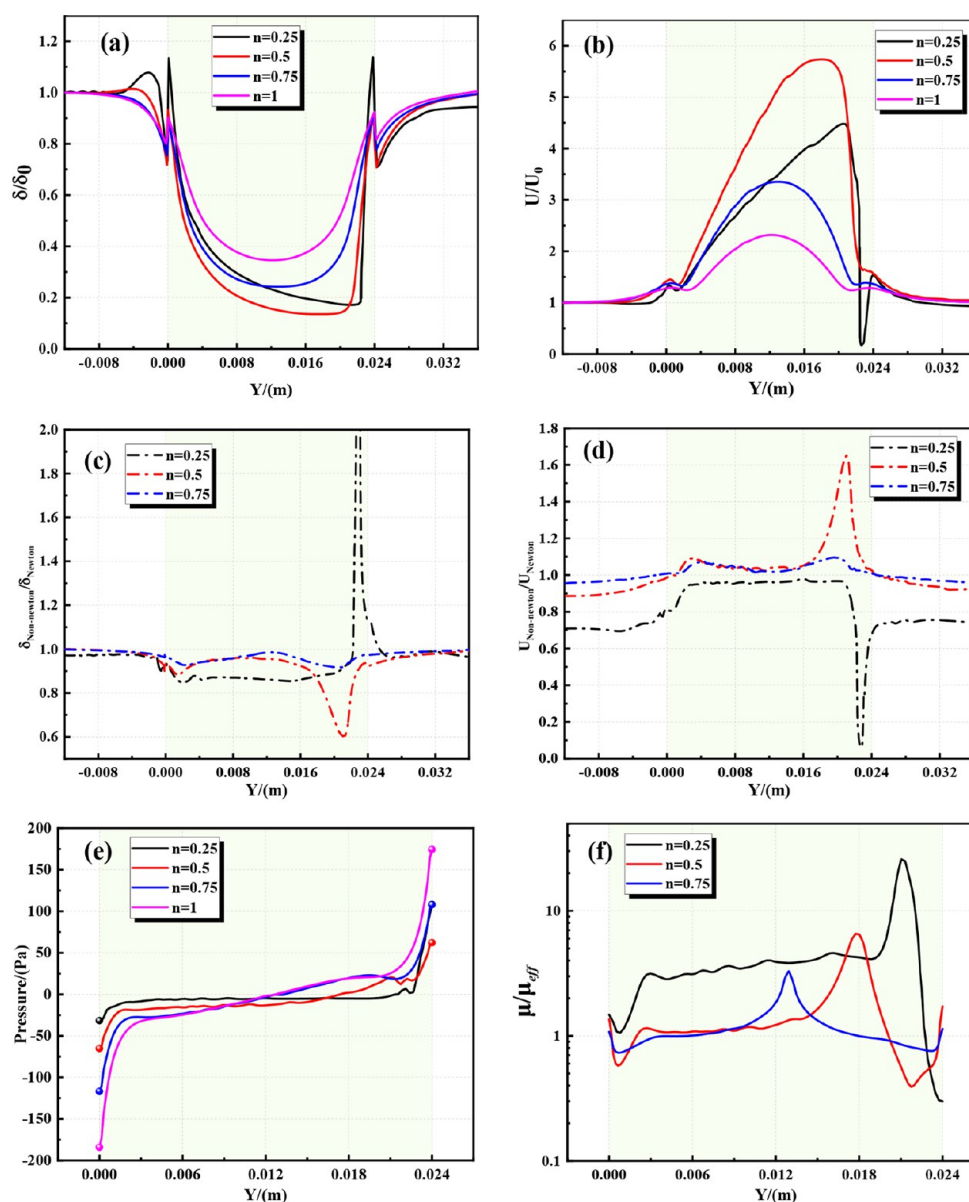


Figure 18. Hydrodynamics of shear-thinning fluid with power law index (n) at $X = 0$. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$; (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$; (e) pressure; and (f) viscosity ($Q = 1.133 \times 10^{-4} \text{ m}^2/\text{s}$, $K = 2$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

with the increase of K . This is because the fluid viscosity increases with higher K , and the deformation of the confined-free film is weakened, so the strain rate remains low. At higher μ_{eff} , the value of $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and $U_{\text{Non-newton}}/U_{\text{Newton}}$ hardly change with the increase of K and are finally symmetrically distributed along the window center. The change in this trend is attributed to the viscosity distribution, as shown in Figures 16f and 17f. Moreover, with the increase of K , the position of the viscosity's extremum moves from the downstream window frame to the window center and is located around the middle of the window.

(2) **Power-Law Index.** The power law index (n) reflects the response of fluid rheology to strain rate. The greater the deviation of n from 1, the more significant the non-Newtonian behavior. Figures 18 and 19 show the hydrodynamics of the non-Newtonian twin-liquid films with different power-law indexes, respectively. For shear-thinning fluids, as n decreases from 1 to 0.25 with the same K , μ_{eff} decreases from 2 to 0.031, and Re_{eff} increases from 0.057 to 3.62. The thinning and acceleration

effects of the confined-free film increase and then decrease, as shown in Figure 18a,b. The reason for this trend is discussed in the previous section, which addresses the dominance of viscous forces or inertial forces. For shear-thickening fluids, the value of μ_{eff} increases with the increase in n . Consequently, the thinning and acceleration effects of the confined-free film also increase and then decrease, as shown in Figure 19a,b.

The conclusion from the previous analysis of the consistency coefficient (K) suggests that the distinction between Newtonian and non-Newtonian fluids diminishes as μ_{eff} increases. Unlike the consistency index K , the power law index n affects not only the value of μ_{eff} but also reflects the sensitivity of the fluid's viscosity to shear stress. For shear-thinning fluids, the combined effect of stronger non-Newtonian properties and lower μ_{eff} makes the values of $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$ and $U_{\text{Non-newton}}/U_{\text{Newton}}$ to deviate from 1 as n decreases. For shear-thickening fluid, as n increases (moving away from 1), μ_{eff} increases, and the difference in film thickness and velocity between non-Newtonian and Newtonian fluid is reduced, similar to the effect of an

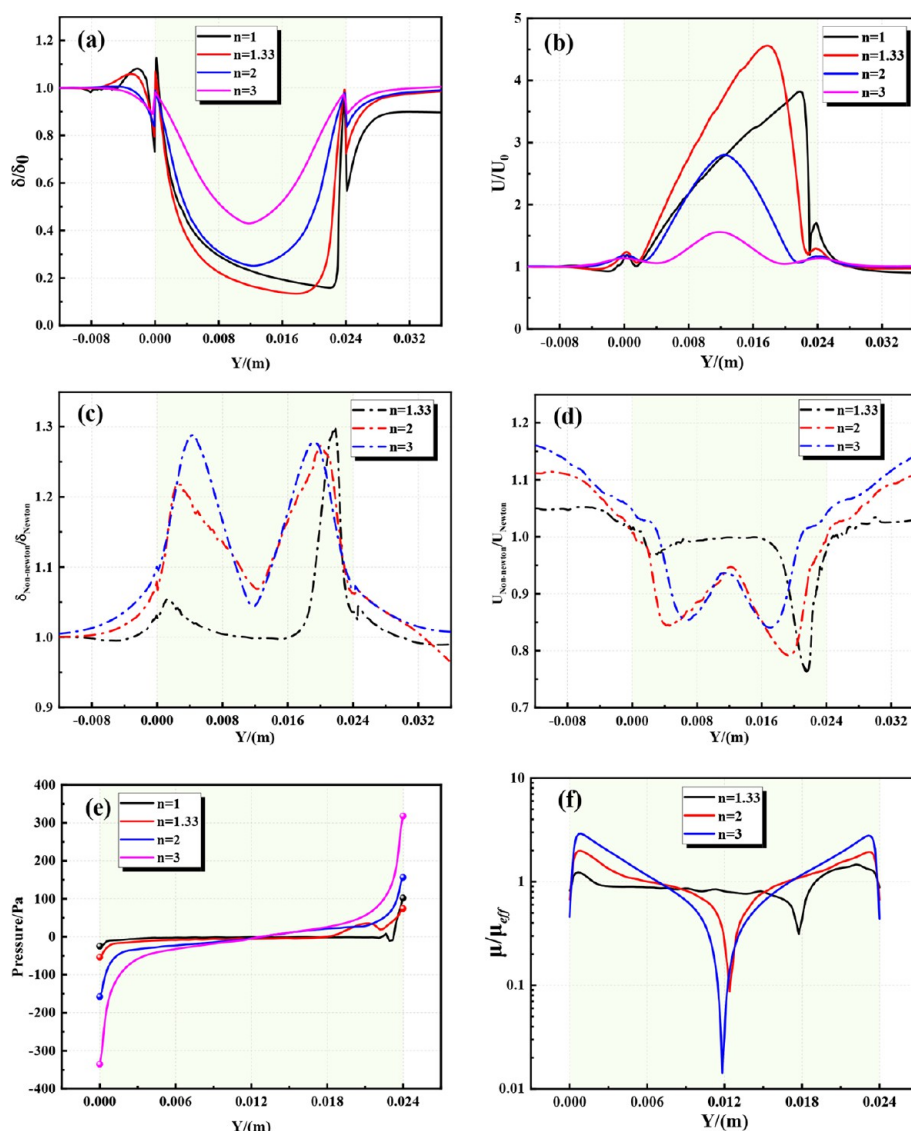


Figure 19. Hydrodynamics of shear-thickening fluid with power law index (n) at $X = 0$. (a) Dimensionless film thickness; (b) dimensionless velocity; (c) $\delta_{\text{Non-newton}}/\delta_{\text{Newton}}$; (d) $U_{\text{Non-newton}}/U_{\text{Newton}}$; (e) pressure; and (f) viscosity ($Q = 1.133 \times 10^{-4} \text{ m}^2/\text{s}$, $K = 1/32$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

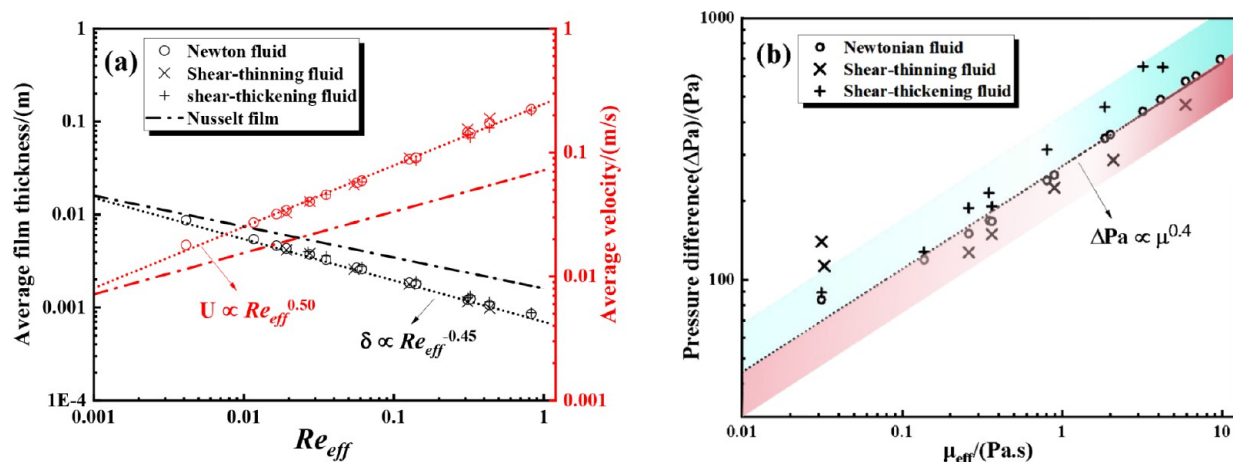


Figure 20. (a) Average film thickness and velocity with different Re ; (b) pressure difference (ΔP) between downstream and upstream window frames with different viscosity ($Q = 1.133 \times 10^{-4} \text{ m}^2/\text{s}$, $W = 24 \text{ mm}$, and $L = 24 \text{ mm}$).

increasing consistency coefficient K . The competitive relationship between stronger non-Newtonian properties and higher μ_{eff}

results in the difference between shear-thickening fluid and Newtonian fluids first increasing and then remaining unchanged

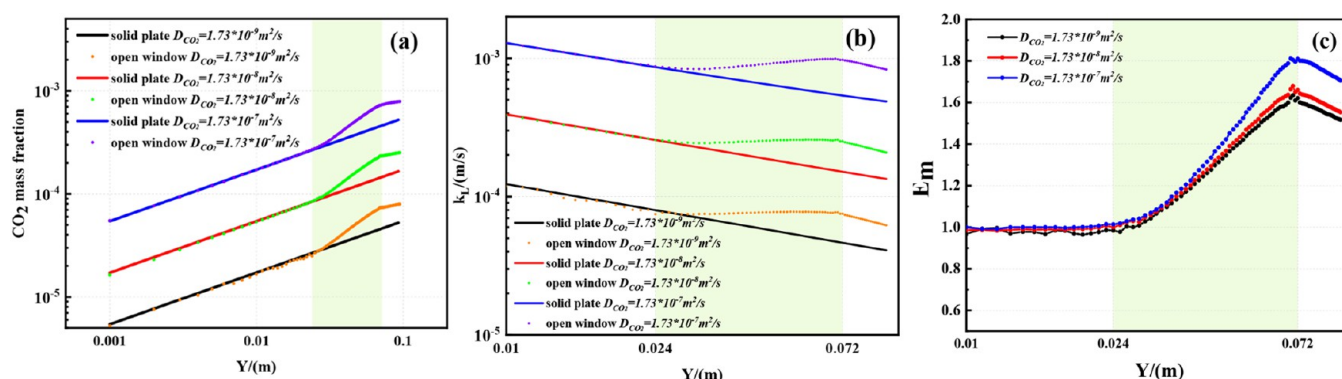


Figure 21. Average CO₂ mass fraction (a), mass transfer rate (b), and enhancing factor (c) of the *xz* plane cross section on the vertical position *Y* with different diffusion coefficient ($\mu_L = 1.256\gamma^{-0.3232}$ Pa·s, $Q = 1.133 \times 10^{-4}$ m²/s, and $Re_{eff} = 0.31$, $W = 24$ mm, and $L = 48$ mm).

as n increases, as shown in Figure 19c,d. Correspondingly, the local viscosity distribution of the non-Newtonian twin-liquid films also changes with variations in n , as shown in Figures 18f and 19f. For shear-thinning fluids, the local viscosity extremum increases as n decreases and gradually transitions toward the downstream window frame. For shear-thickening fluids, the local viscosity extremum increases with the increase of n and gradually transitions toward the window center.

4.2.3. Critical Hydrodynamic Factor Analysis. The influence of the discussed rheological properties (i.e., K and n) and flow rate Q on the confined-free film can be considered comprehensively using the effective Reynolds number, Re_{eff} , which is defined as eq 12. Figure 20a shows the relationship between Re_{eff} (0.001–1) and the average film thickness and velocity of the confined-free film under the same window size. Interestingly, the average film thickness and velocity of the non-Newtonian fluid and Newtonian fluid with the same Re_{eff} are very close overall. Compared with the Nusselt film, both confined-free films have thinner thickness and higher velocity, which are beneficial to interfacial mass transfer. This is consistent with our previous work on Newtonian fluids.^{2,35} Moreover, different from the Nusselt film, in which the thickness and velocity are proportional to $-1/3$ and $1/3$ of Re_{eff} respectively, the confined-free film's thickness and velocity are proportional to -0.45 and 0.5 of Re_{eff} . This means that the confined-free film is more sensitive to Re_{eff} rather than a specific rheological behavior.

As discussed in Section 4.1, the dominant force for the confined free film changes from viscous force to inertial force with the increase in window length. Longer windows can reduce the adverse pressure gradients $\frac{\partial \Delta P}{\partial Y}$ between the upstream and downstream window frames, which is conducive to the confined free film's acceleration under inertial force. Figure 20b shows the relationship between ΔP and μ_{eff} with the same window size. For Newtonian fluids, the pressure difference ΔP is proportional to $\mu^{0.4}$. ΔP has a negative deviation of about 10% for shear-thinning fluid and a 10% positive deviation for shear-thickening fluid compared with Newtonian fluids. Accordingly, the window length is suggested to be proportional to $0.9 \mu_{eff}^{0.4}$ for shear-thinning fluid and $1.1 \mu_{eff}^{0.4}$ for shear-thickening fluid. It is worth noting that when μ_{eff} is less than 0.1 Pa·s, Re_{eff} exceeds 1. The confined-free film within the window is dominated by inertial forces rather than viscous forces. The maximum liquid film velocity shifts from the center of the window to its lower edge. Upon impacting the lower edge of the window, a greater proportion of the inertial momentum is converted into

hydrostatic pressure, resulting in an additional pressure increase at the lower edge of the window.

The results in Sections 4.1 and 4.2 have confirmed that the rheological properties of the fluid have a significant influence on the specific hydrodynamic characteristics of the confined-free film. However, the results in Figure 20 indicate a general rule that, regardless of the fluid's non-Newtonian behavior—shear thickening or shear thinning—the average hydrodynamic characteristics of the confined-free film can be effectively characterized by Re_{eff} and μ_{eff} . This result not only reveals a hydrodynamic commonality of the confined-free film but also guides the optimization design of the opening window size from the perspective of practical application. We speculate that this rule may also apply to the wider free flow, but it needs further experimental verification to confirm.

4.3. Mass Transfer. The significance of the twin-liquid film is reflected in its interfacial mass transfer performance. For the sake of clarity, the mass transfer rate of gas in a liquid is described by the liquid-side mass transfer coefficient (k_L). According to the mass balance on the total falling film:

$$k_L(C^* - C) dY = Q dC \quad (32)$$

The boundary conditions are

$$\begin{aligned} Y = 0: & \quad C = 0 \\ Y = L: & \quad C = C_{out} \end{aligned} \quad (33)$$

Thus, eq 34 can be integrated to yield the total liquid-side mass transfer coefficient as

$$k_L = \frac{Q}{L} \ln \frac{C^*}{C^* - C_{out}} \quad (34)$$

We introduce a mass transfer enhancement factor E_m as the ratio of the mass transfer rate between the twin-liquid film with the opening window to a theoretical one for the wall-bound film on a solid plate:

$$E_m = \frac{k_L}{k_W} \quad (35)$$

where k_L is the total liquid-side mass transfer coefficient for the opening window plate and k_W is the coefficient for the solid plate.

The mass transfer of gas in a liquid mainly arises from two contributions: convective mass transfer and diffusion. The convective mass transfer is related to the hydrodynamics of liquid films, as discussed in Section 4.2. Diffusion mainly affects

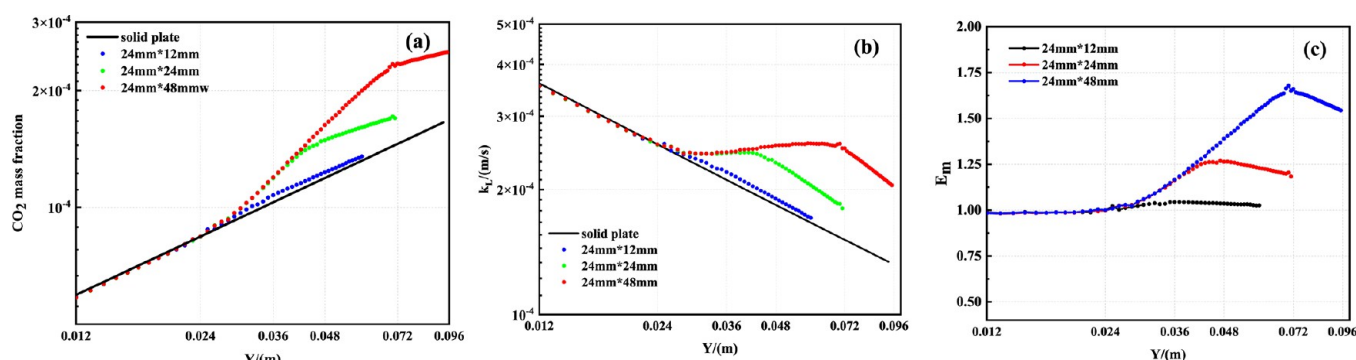


Figure 22. Average CO₂ mass fraction (a), mass transfer rate (b), and enhancing factor (c) of the xz plane cross section on the vertical position Y with different window length ($\mu_L = 1.256\gamma^{-0.3232}$ Pa·s, $Q = 1.133 \times 10^{-4}$ m²/s, and $Re_{eff} = 0.31$, $D_{CO_2} = 1.73 \times 10^{-8}$ m²/s).

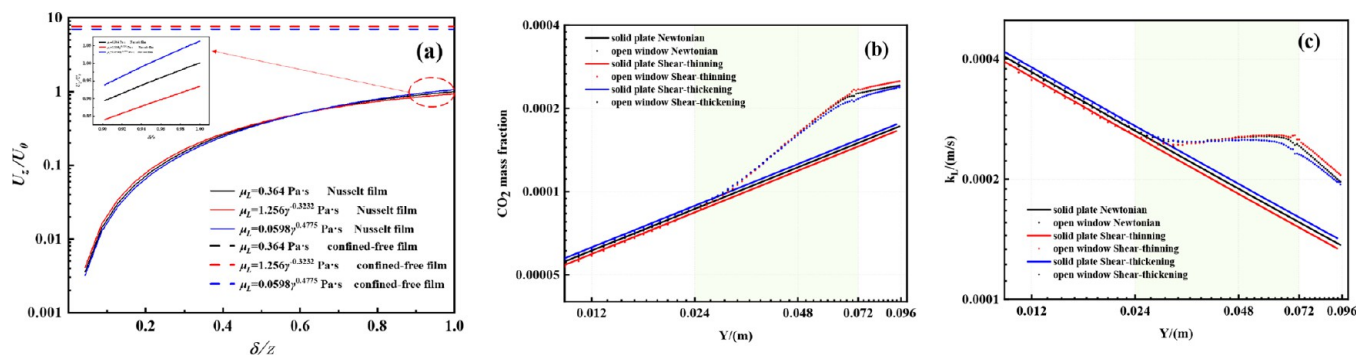


Figure 23. Average CO₂ mass fraction (a), mass transfer rate (b) of the xz plane cross section on the vertical position Y with different rheological properties fluid; (c) dimensional liquid film velocity distribution along the z direction at $y = 0.024$ m ($\mu_L = 0.364$ Pa·s for Newtonian fluid, $\mu_L = 1.256\gamma^{-0.3232}$ Pa·s for shear-thinning fluid, $\mu_L = 0.0598\gamma^{0.4775}$ Pa·s for shear-thickening fluid, $Re_{eff} = 0.311$, $D_{CO_2} = 1.73 \times 10^{-8}$ m²/s, $W = 24$ mm, and $L = 48$ mm).

the mass transfer boundary layer and should also be taken seriously. Convection and diffusion work together to influence the mass transfer rate. The diffusion ability of a gas in a liquid is characterized by the diffusion coefficient. For most gases in Newtonian fluids, their diffusion coefficient is usually constant due to constant viscosity. However, for some non-Newtonian fluids, the diffusion coefficient of gas in non-Newtonian fluids is not constant. According to the experimental results from the work of Wasan et al.,²⁷ the diffusivity in many non-Newtonian systems (e.g., CMC or Polyox solutions) is a function of viscosity. To investigate the effect of diffusion on the mass transfer performance of non-Newtonian twin-liquid films, the cases in which the diffusion coefficient of CO₂ is independent of liquid local viscosity, as well as those in which it depends on liquid local viscosity, are considered, respectively, in Sections 4.3.1 and 4.3.2.

4.3.1. Diffusion Coefficient Independent of Viscosity. Taking shear-thinning fluid as an example, it is assumed that the diffusion coefficient does not change with the change in liquid viscosity; the window size is 24 mm × 48 mm. The effect of diffusion coefficient D_{CO_2} on mass transfer enhancement of the twin-liquid film has been investigated, as shown in Figure 21. As investigated by Albert,³⁶ three artificial diffusion coefficients, 1.73×10^{-9} , 1.73×10^{-8} , and 1.73×10^{-7} m²/s, were chosen, and the Sc numbers were 2×10^5 , 2×10^4 , and 2×10^3 .

It can be observed that the k_L is the same as that of the solid plate on the upper wall region ($Y < 0.024$ m), and it increases significantly by 75% in the window region (0.024 m $< Y < 0.048$ m), as shown in Figure 21. On the lower wall ($Y > 0.048$ m), the

decreasing trend of k_L is consistent with that of the solid plate. According to Xie's result,² higher flow velocity enhances streamwise convective mass transfer, and the inner concentration boundary layer can be stretched with decreasing film thickness. With an increase in D_{CO_2} , the Schmidt numbers decrease. In laminar boundary layer mass transfer, the concentration boundary layer thickness is proportional to $Sc^{-1/3}$, resulting in a thinner concentration boundary layer, which is conducive to mass transfer. Meanwhile, the confined-free film accelerates, and its thickness is reduced. The former enhances the convective mass transfer, and the latter shortens the diffusion distance. Therefore, the enhancement effect of the opening window is more significant with higher D_{CO_2} , as shown by E_m increasing for $D_{CO_2} = 1.73 \times 10^{-7}$ in Figure 22c.

On the other hand, under the same diffusion coefficient, enhancing convective mass transfer can significantly increase the whole mass transfer rate. Figure 22 shows the values of k_L and E_m along the y direction with different window lengths; the window width is consistently 24 mm. As the window length increases from 12 mm to 48 mm, the value of E_m increases from 1 to 1.6. This is attributed to the higher film velocity and smaller film thickness associated with a longer window length, as revealed in Figure 11. The synergistic enhancement of intensified convective transfer and diffusion promotes interfacial mass transfer by the twin-liquid film.

Figure 23a and b show the average CO₂ mass fraction and mass transfer rate distribution along the y direction with different rheological properties of the fluid. All the fluids have the same μ_{eff} and Re_{eff} and the diffusion coefficient D_{CO_2} is

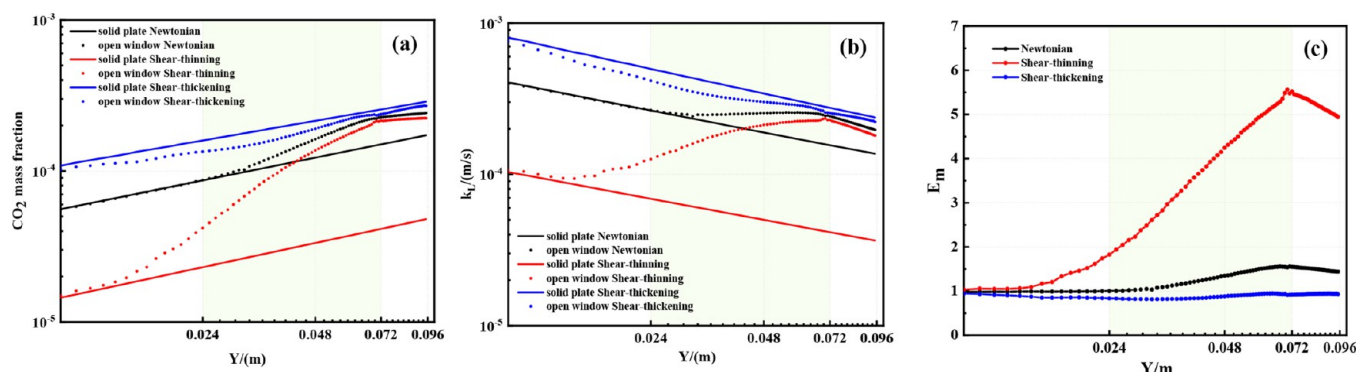


Figure 24. Average CO₂ mass fraction (a), mass transfer rate (b), and enhancing factor (c) of the xz plane cross section on the vertical position y with different rheological properties fluid. ($\mu_L = 0.364$ Pa·s for Newtonian fluid, $\mu_L = 1.256\gamma^{-0.3232}$ Pa·s for shear-thinning fluid, $\mu_L = 0.0598\gamma^{0.4775}$ Pa·s for shear-thickening fluid, $Re_{eff} = 0.31$, $\mu_{eff} = 0.364$ Pa·s, $D_{CO_2} = 0.63 \times 10^{-8}$ m²/s, $W = 24$ mm, and $L = 48$ mm).

assumed to be 1.73×10^{-8} m²/s. As shown in Figure 23a, for the Nusselt wall-bounded film on a solid plate ($Y < 0.024$ m), the parabolic film velocity distribution for different rheologies along the Z direction (direction of film thickness) has subtle differences. It can also be noticed that the velocity distribution is uniform within the film. Besides, as shown in Figure 8ab, the confined-free film's thickness and velocity of the Newtonian and non-Newtonian fluids with the same μ_{eff} and Re_{eff} are similar before reaching the downstream window frame. Hence, the mass transfer rate distributions of the three fluids almost overlap, as shown in Figure 23c. Until the downstream window frame is reached, there is a slight difference in the mass transfer rate of the three fluids. This is attributed to the differences in the local flow field, as discussed in Section 4.2.

4.3.2. Diffusion Coefficient Dependent on Viscosity. According to the Stokes–Einstein equation, the diffusion coefficient is proportional to μ^{-1} for some solution. Assuming that the diffusion coefficient is 1.73×10^{-8} m²/s while the liquid viscosity is 0.364 Pa·s. So the relationship between the diffusion coefficient and the liquid local viscosity is

$$\frac{D_{CO_2}}{1.73 \times 10^{-8}} \sim \frac{0.364}{\mu} \quad (36)$$

Figure 24 shows the mass transfer performance of the twin-liquid film with the diffusion coefficient D_{CO_2} being dependent on viscosity. For the Nusselt wall-bounded film on a solid plate, all of the fluids have the same film thickness and almost identical surface velocity. Due to the very small shear stress near the gas–liquid interface, the viscosity of shear-thickening fluid near the interface is far lower than the μ_{eff} and viscosity of shear-thinning fluid is far higher than μ_{eff} . Correspondingly, the shear-thickening fluid had the highest diffusion coefficient. A higher diffusion coefficient is beneficial for improving diffusion, thereby increasing the interfacial mass transfer rate, as discussed in Section 4.3.1. Thus, the k_L of shear-thickening fluid is the highest, followed by Newtonian fluid, with shear-thinning fluid being the lowest, as shown in Figure 24b.

It can also be noticed that k_L slightly decreases compared with that on the solid plate for shear-thickening, as shown in Figure 24b. For shear-thickening fluids, due to the stretching and contraction of the confined-free film inside the window, the shear stress near the liquid film interface is increased, and the local viscosity for the confined-free film is higher than that for the wall-bounded film. Consequently, the diffusion coefficient decreases. Therefore, despite the confined-free film's thinning

and acceleration within the opening window, k_L has a slight decrease. Conversely, for shear-thinning fluids, the stretching and contraction of the confined-free film increase the shear rate and reduce the local viscosity, thereby improving the diffusion coefficient significantly. This not only improves convective mass transfer but also enhances diffusion. As a result, the interfacial mass transfer rate is enhanced significantly, as shown in Figure 24c.

Comparatively, E_m is 1.75 with the diffusion coefficient independent of viscosity, and E_m increases to 5.5 with the diffusion coefficient dependent on viscosity for shear-thinning fluids, as shown in Figures 22c and 24c. Therefore, the opening window can provide synergistic enhancement for the shear-thinning fluid with its diffusion coefficient dependent on viscosity.

5. CONCLUSION

The twin-liquid film, which consists of a wall-bounded film supported by a solid wall and confined-free film through the opening window, demonstrates excellent mass transfer performance for high-viscosity Newtonian fluids. In this work, the application of the twin-liquid film has been expanded to non-Newtonian fluid.

For non-Newtonian and Newtonian fluids with the same μ_{eff} and Re_{eff} , the confined-free film can achieve the same average liquid film thickness and velocity for a certain window size. The stretching and contraction of the confined-free film lead to viscosity distributions near the window frame that are opposite for shear-thickening fluids and shear-thinning fluids. Thereby, the local film thickness and velocity distribution near the window frame differ. The local differences in film thickness and velocity between the non-Newtonian and Newtonian fluids increase with the window length and plate thickness. In addition, as μ_{eff} increases, the confined-free film transitions from inertial-dominated flow to viscous-dominated flow, and the thinning and acceleration effects of the liquid film first increase and then decrease. For viscosity-dominated flow, the local viscosity distribution remains symmetric about the window's center and is invariant to increases in the consistency coefficient K . It is suggested that the window length is proportional to $\mu^{0.4}$ for Newtonian fluid, $0.9 \mu_{eff}^{0.4}$ for shear-thinning fluid, and $1.1 \mu_{eff}^{0.4}$ for shear-thickening fluid.

For the case of the diffusion coefficient independent of viscosity, the average hydrodynamic characteristics of the Newtonian and non-Newtonian confined-free film with the same μ_{eff} and Re_{eff} are almost identical, resulting in similar mass

transfer characteristics. Compared to the solid plate, the mass transfer rate of the plate with 24 mm × 48 mm windows increases by 75%. When the diffusion coefficient is inversely proportional to viscosity, for shear-thinning fluids, the stretching and contraction of the confined-free film enhance shear rates, reducing local viscosity and substantially enhancing the diffusion coefficient. This dynamic behavior of the shear-thinning film not only boosts convective mass transfer but also amplifies diffusion, synergistically enhancing the mass transfer rate. The results indicate that the mass transfer rate of the plate with 24 mm × 48 mm windows increases by 5.5 times compared to the solid plate for the shear-thinning fluid when its diffusion coefficient is inversely proportional to viscosity.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

■ AUTHOR INFORMATION

Corresponding Author

Yuan Zong – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0000-0001-6410-3904; Email: zongyuan@ecust.edu.cn

Authors

Long He – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0009-0004-5613-8401

Hanguang Xie – Hydrogen Safety Engineering and Research Centre (HySAFER), Ulster University, Newtownabbey, Northern Ireland BT37 0QB, U.K.

Zhenhua Yao – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China

Ziming Dong – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China

Ling Zhao – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China; orcid.org/0000-0001-5239-1152

Gance Dai – State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.iecr.5c01410>

Author Contributions

L.H.: investigation, software, writing—original draft, and date curation; H.X.: investigation; Z.Y.: investigation; Z.D.: investigation; Y.Z.: conceptualization, methodology, writing—review and editing, and supervision; L.Z.: investigation and resources; G.D.: conceptualization and methodology

Notes

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■ NOMENCLATURE

a	Effective mass transfer area (m^2)
B_{CO_2}	CO_2 mass flow rate (kg/s)
B_{liquid}	liquid mass flow rate (kg/s)
C_{CO_2}	CO_2 concentration in absorbent
$C_{\text{CO}_2}^*$	CO_2 equilibrium concentration in absorbent
D_{CO_2}	diffusion coefficient of CO_2 in absorbent (m^2/s)
$\bar{D}$	rate-of-deformation tensor
E_m	ratio between real mass transfer rate and the theoretical value of a solid plate
F_{vol}	surface tension source term
g	gravitational acceleration (m/s^2)
K	consistency coefficient
k_L	mass transfer coefficient (m/s)
k_W	theoretical total liquid-side mass transfer coefficient for solid plate (m/s)
L	window length (m)
M	the number of grids in a film thickness
$\mathbf{n}$	normal vector of the interface
n	power law index
P	pressure (Pa)
ΔP	pressure difference (Pa)
Q	volume flow rate per unit plate width (m^2/s)
R	the ratio of grid height and length
Re_{eff}	effective Reynolds number
Sc	Schmidt number
T	plate thickness (m)
U	velocity magnitude (m/s)
$\mathbf{u}$	fluid velocity vector (m/s)
u	velocity in x direction (m/s)
v	velocity in y direction (m/s)
w	velocity in z direction (m/s)
W	window width (m)
X	X coordinate (m)
Y	Y coordinate (m)
Z	Z coordinate (m)

■ GREEK LETTERS

μ_L	liquid viscosity ($\text{Pa}\cdot\text{s}$)
μ_G	gas viscosity ($\text{Pa}\cdot\text{s}$)
μ_{eff}	effective viscosity ($\text{Pa}\cdot\text{s}$)
ρ_L	liquid density (kg/m^3)
ρ_G	gas density (kg/m^3)
α	volume fraction
β	standard deviation
δ	film thickness (m)
σ	surface tension (N/m)
τ	shear stress
γ	shear rate
ν	kinematic viscosity (m^2/s)
κ	gas–liquid interface curvature ($1/\text{m}$)
λ	friction coefficient

■ ABBREVIATIONS

CCS	carbon capture and storage
CMC	carboxymethyl cellulose
CSF	continuum surface force
ILs	ionic liquids
RDR	rotating disk reactor
RPB	rotating packed bed
VOF	volume of fluid

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